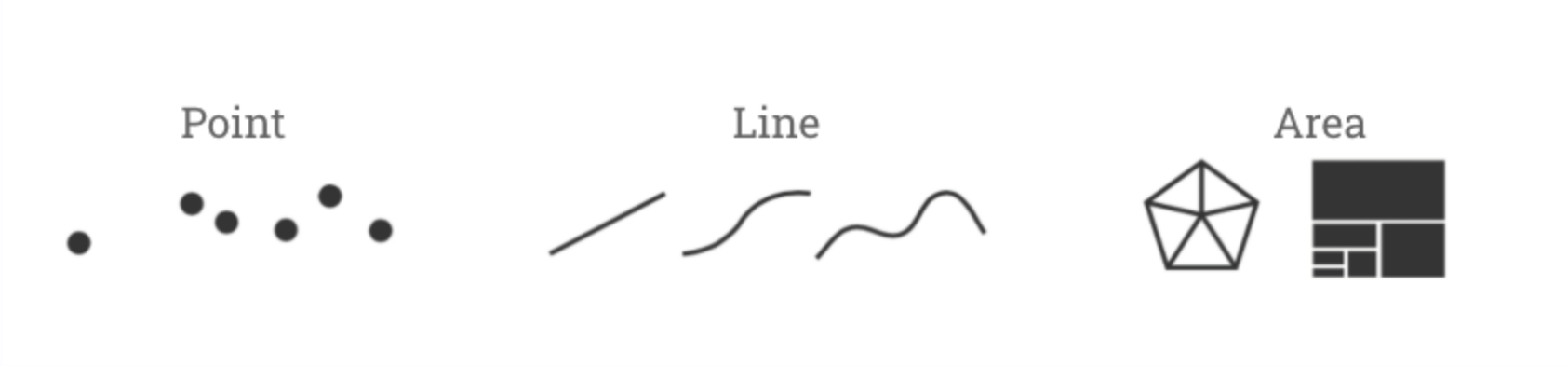
# Language of Data Visualization

# Marks and Channels



# Marks ・　マーク

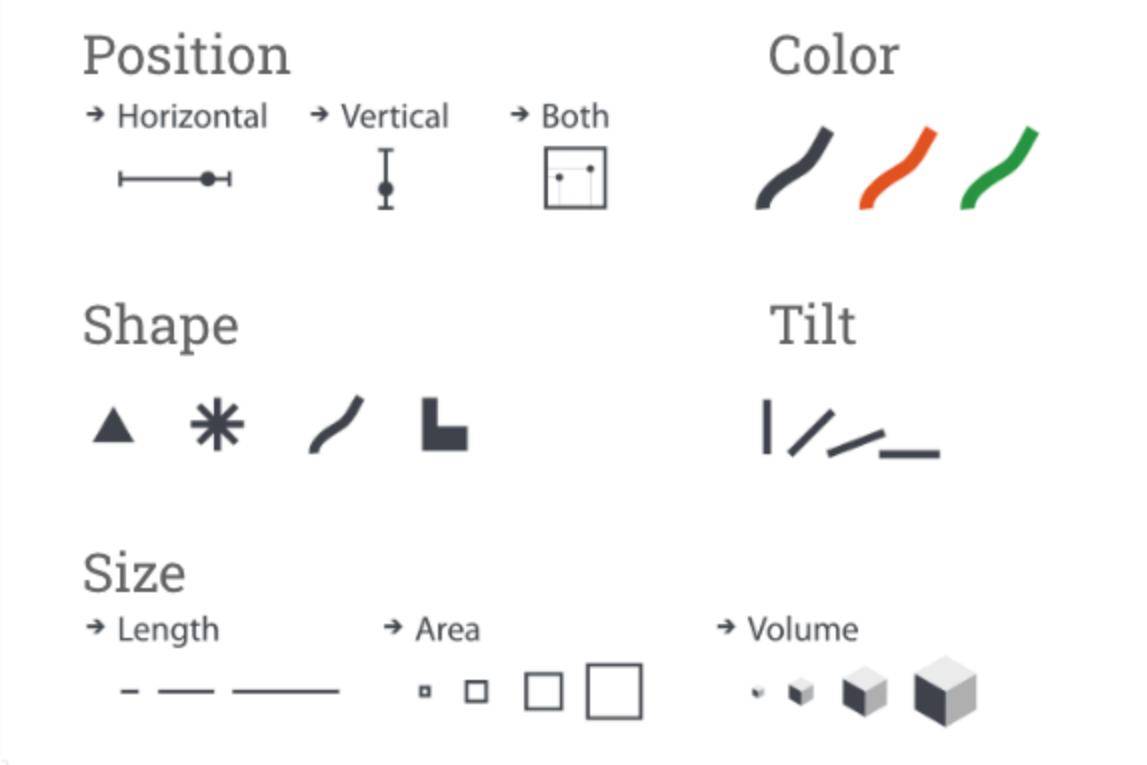
The shapes that represent data.
データを表す形




Graphics from Munzner. Visualization Analysis and Design (2015)

# Channels ・ チャンネル

Ways to vary the appearance of marks
マークの見た目を変える方法



# What does a circle represent?

Gapminder: one country, one year



Circle

A mark

# What does a circle represent?

Gapminder: one country, one year



Circle

A mark

Area

Population

# What does a circle represent?

Gapminder: one country, one year



| Circle | Area | Color |
| --- | --- | --- |
| A mark | Population | Continent |

# What does a circle represent?

Gapminder: one country, one year



Circle

A mark



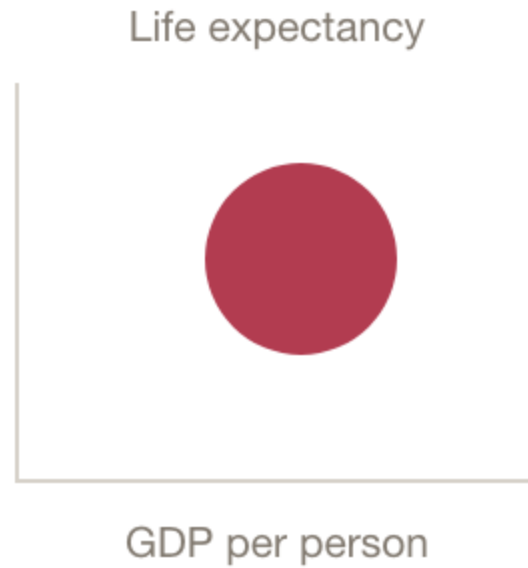
Area

Population



Color

Continent

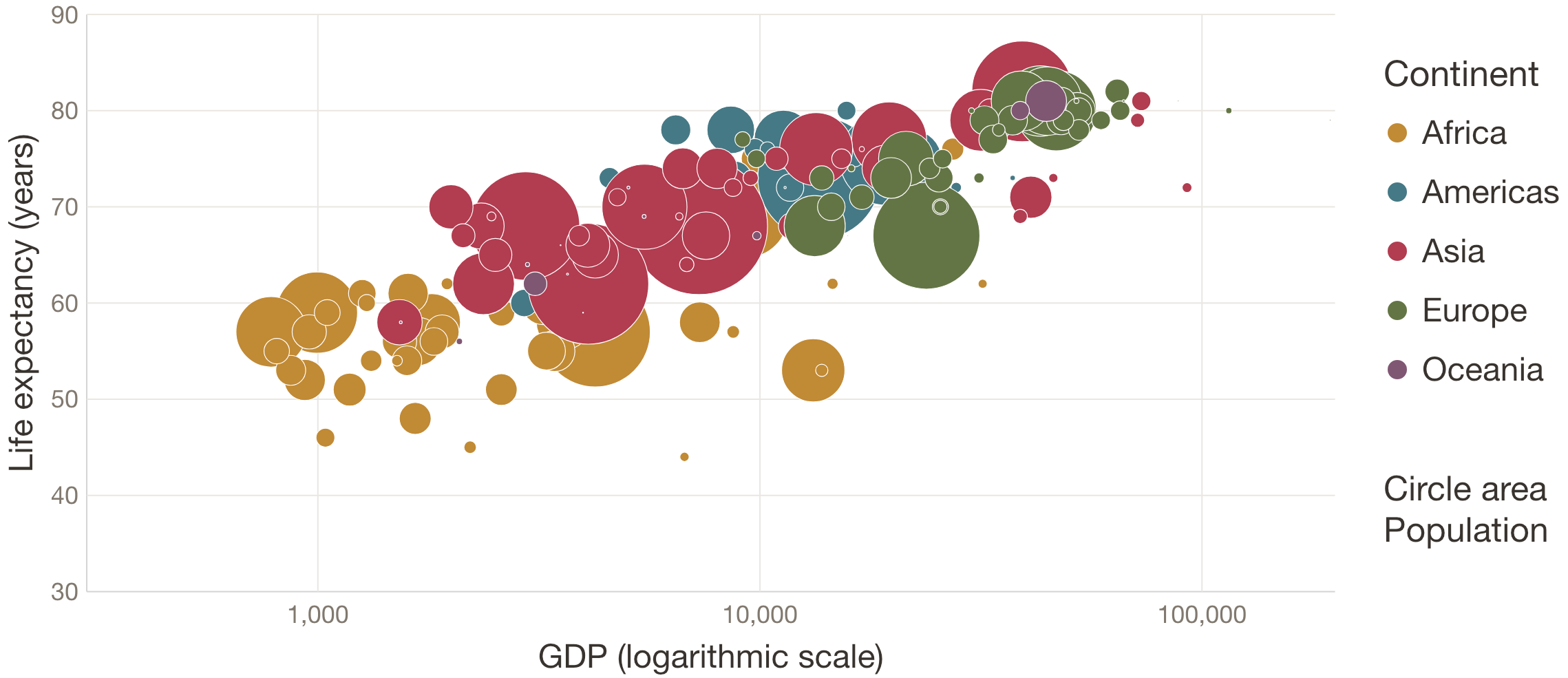



Position

GDP + life expectancy

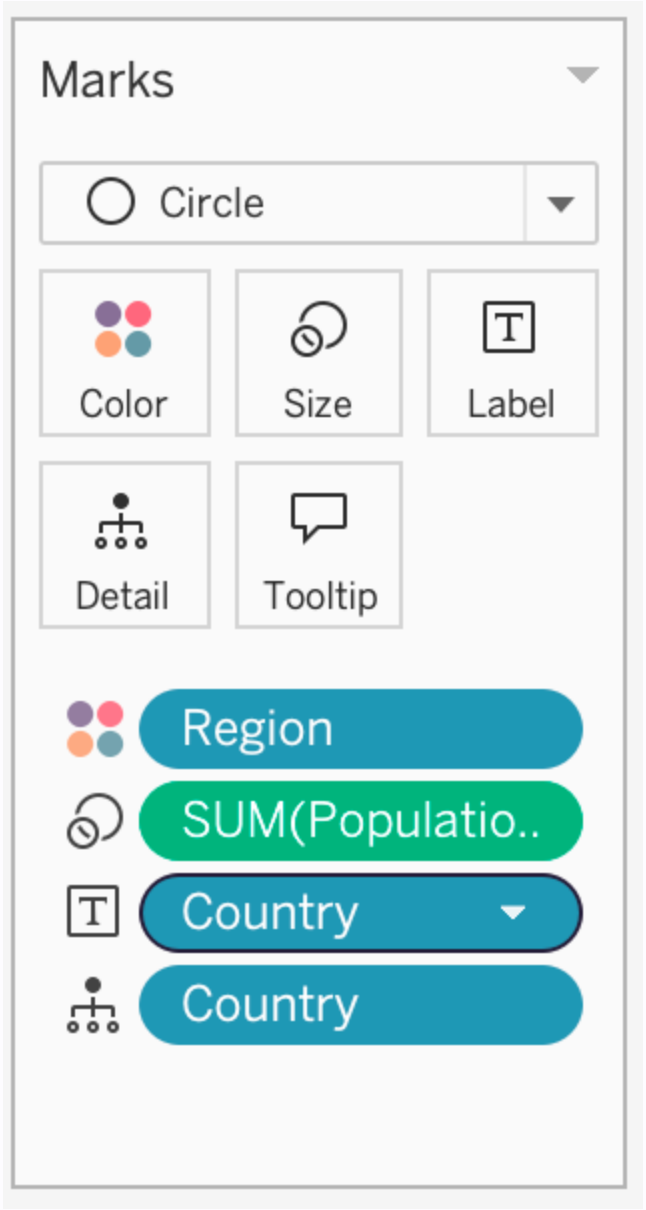
# Gapminder: the channels together

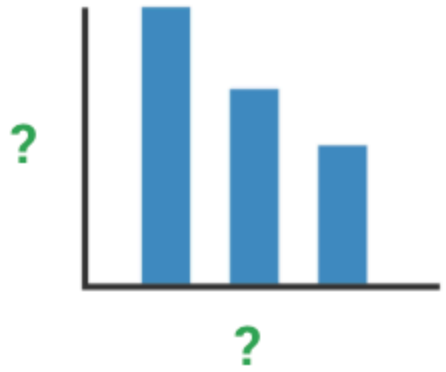
2007



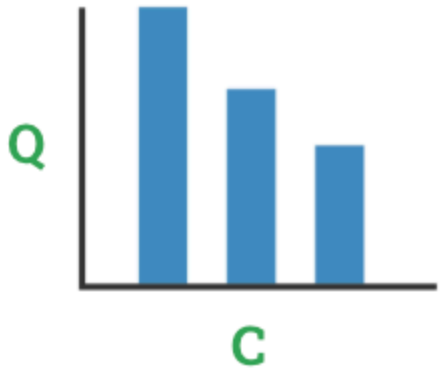

One circle = one country

175 complete records; 22 omitted for missing / invalid fields.

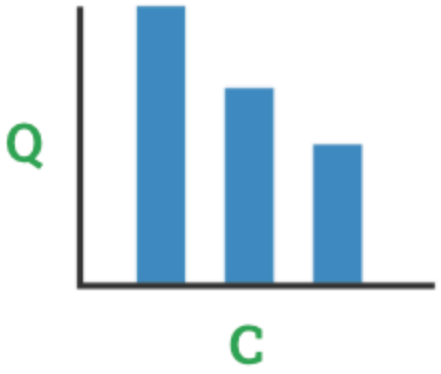

Marks
Circle
Color
Size
Label
Detail
Tooltip
Region
SUM(Populatio..
Country
Country



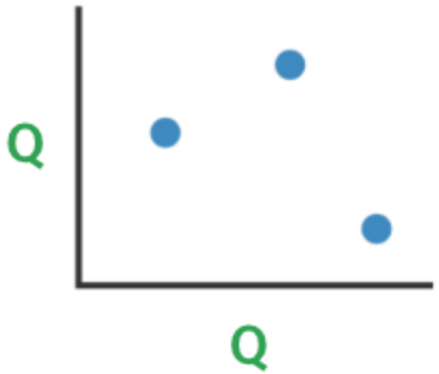

Mark: ?



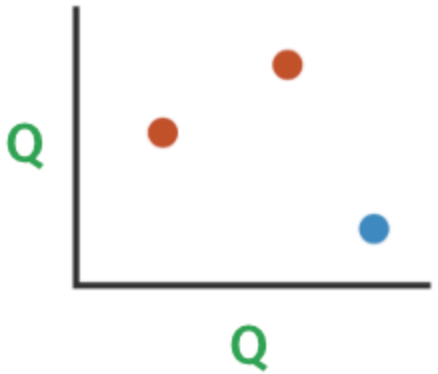

Mark: **line**



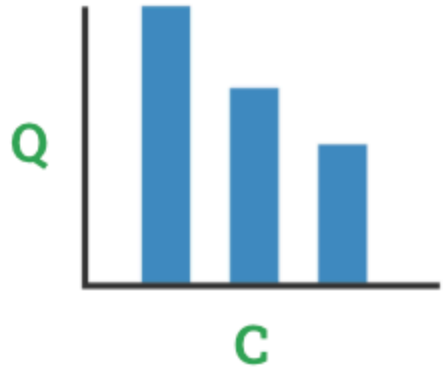

Mark: **line**



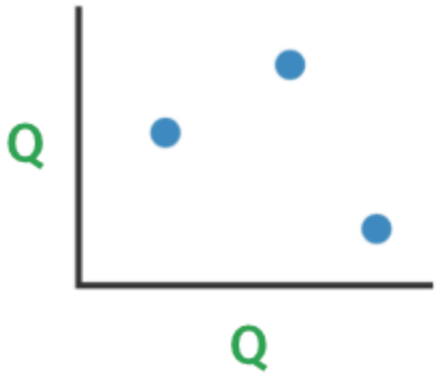

Mark: **point**



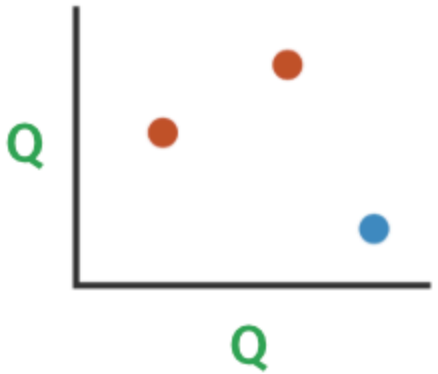

Mark: **point**
Channels:
Color: **C**



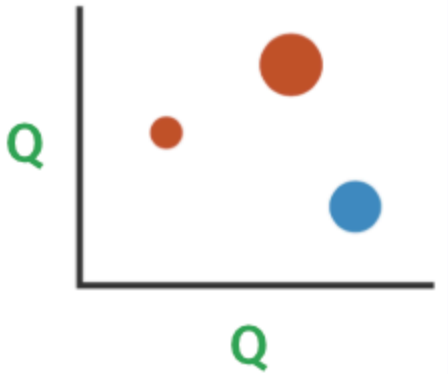

Mark: **line**



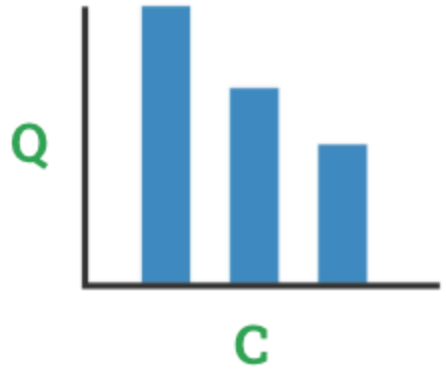

Mark: **point**



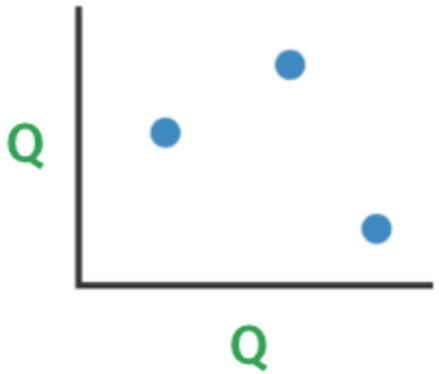

Mark: **point**
Channels:
Color: **C**



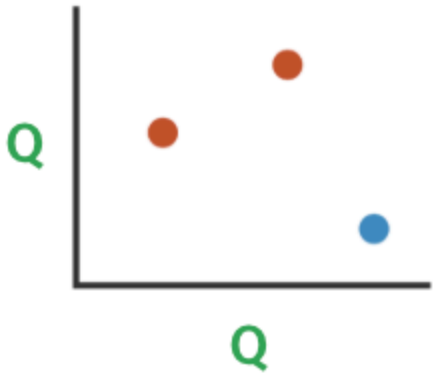

Mark: **point**
Channels: **?**



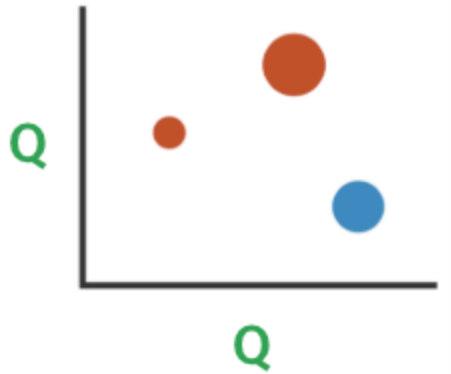

Mark: **line**




Mark: **point**




Mark: **point**
Channels:
Color: **C**




Mark: **point**
Channels:
Color: **C**
Size: **Q**

## Position

→ Horizontal → Vertical → Both



## Color



## Shape



## Tilt



## Size

→ Length → Area → Volume



Position

→ Horizontal → Vertical → Both



Color



Categorical



Shape



Tilt



×

Ordinal



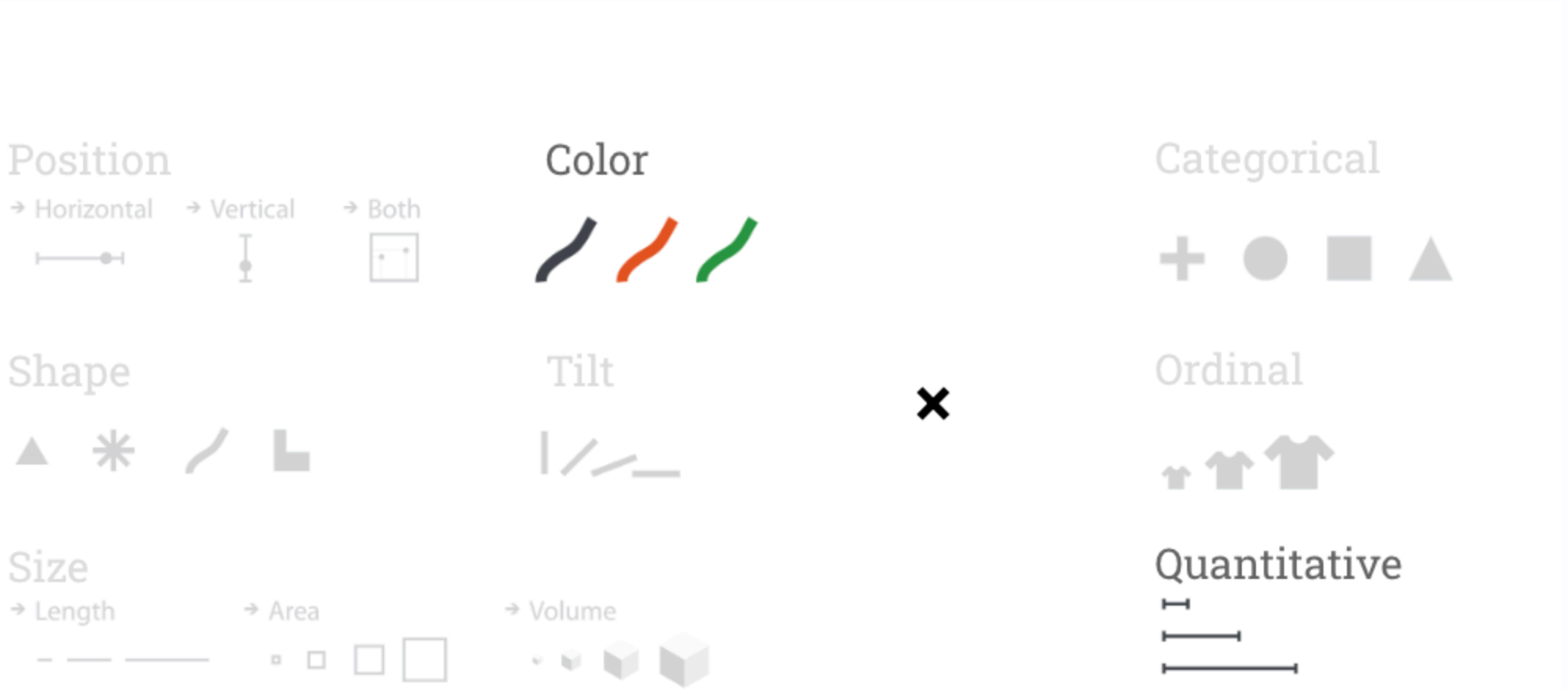
Size

→ Length → Area → Volume



Quantitative




Position
Horizontal
Vertical
Both
Color
Categorical
Shape
Tilt
Ordinal
Size
Length
Area
Volume
Quantitative

Filters

Marks

Circle

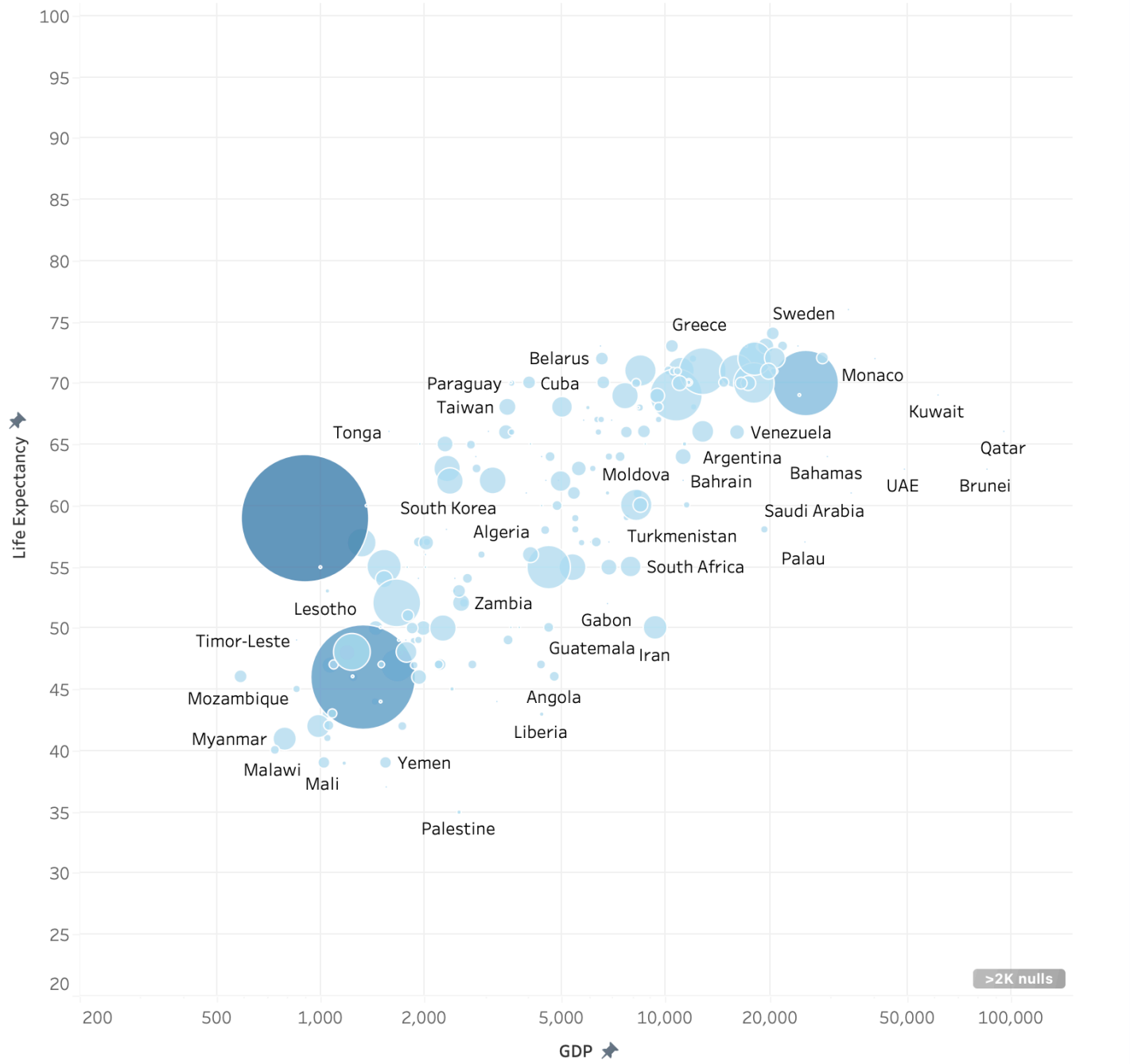
Color | Size | Label | Detail | Tooltip

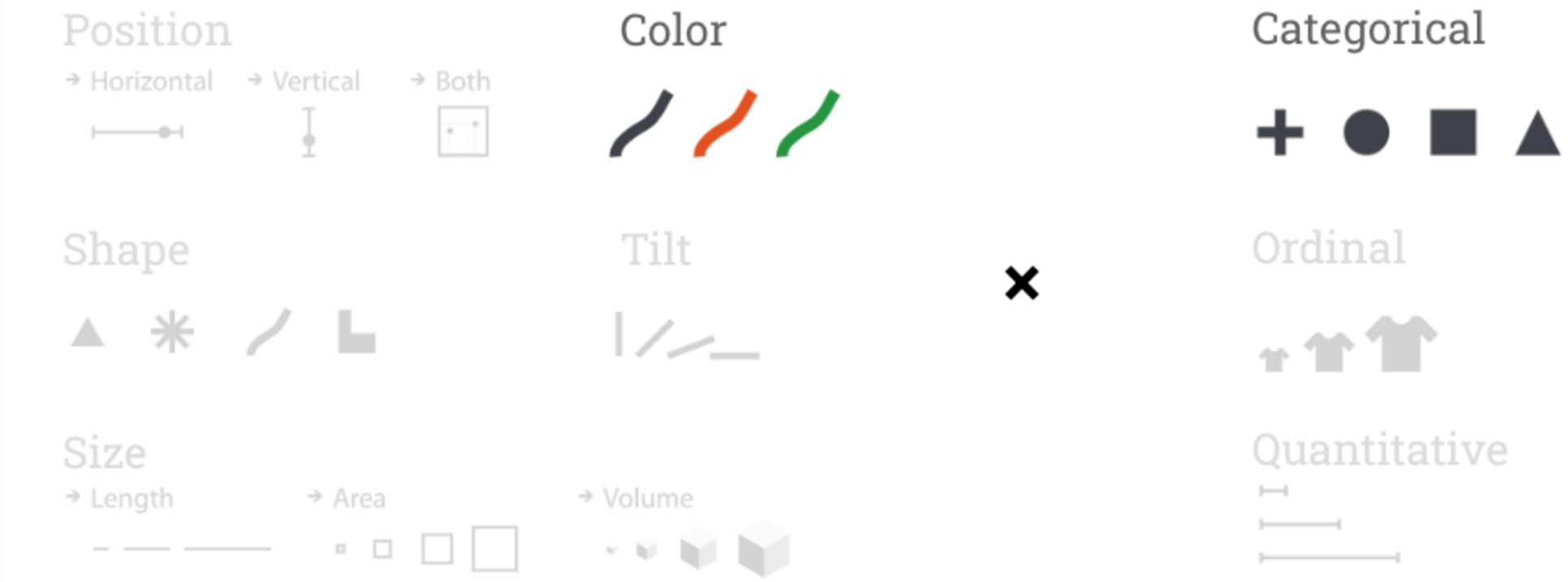
SUM(Populatio..
SUM(Populatio..
Country
Country

SUM(Population)

651 999M

# Hans - 1967




Position
Horizontal
Vertical
Both
Color
Categorical
Shape
Tilt
Ordinal
Size
Length
Area
Volume
Quantitative

Filters

Marks

Circle

Color | Size | Label

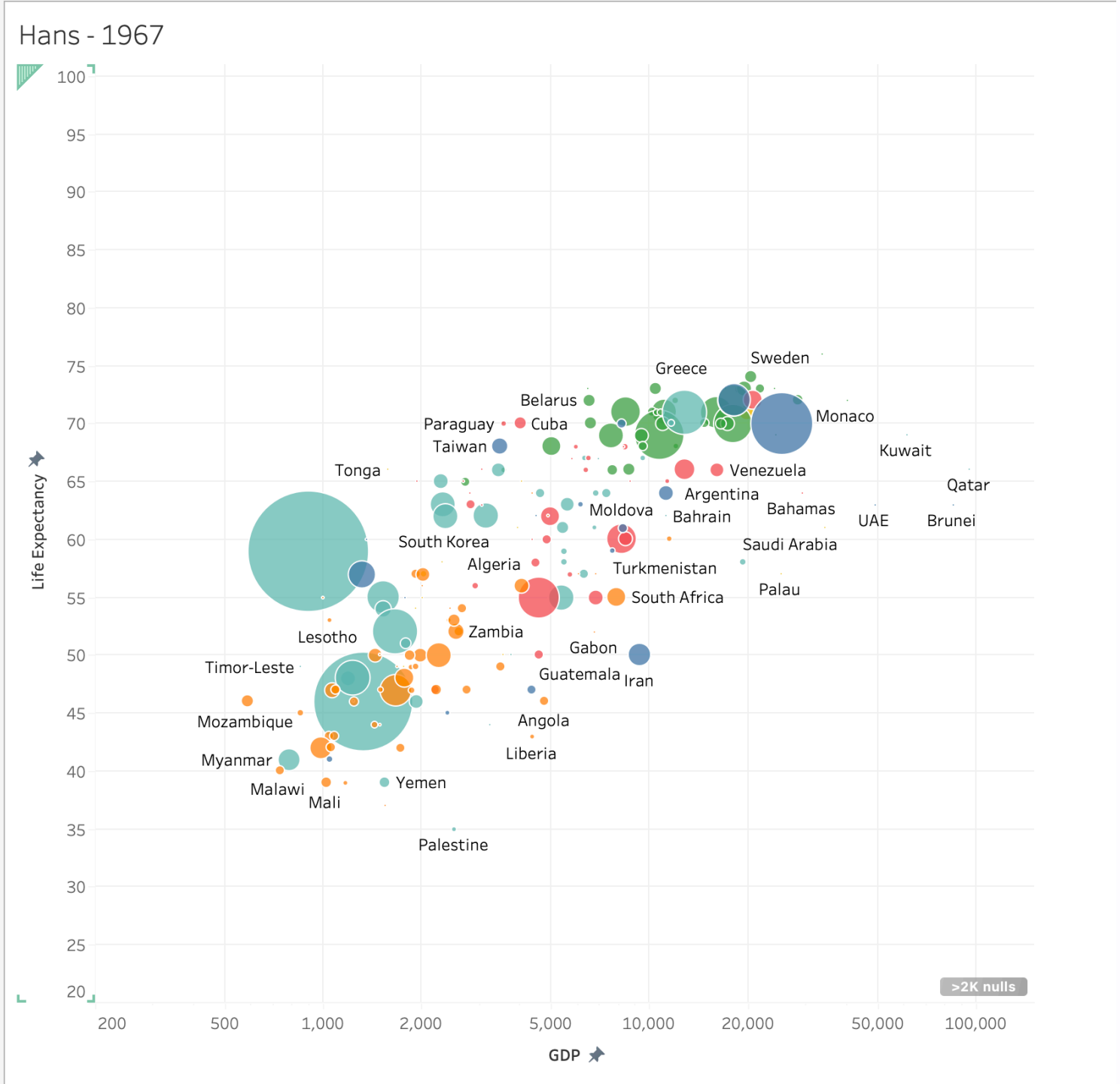
Detail | Tooltip

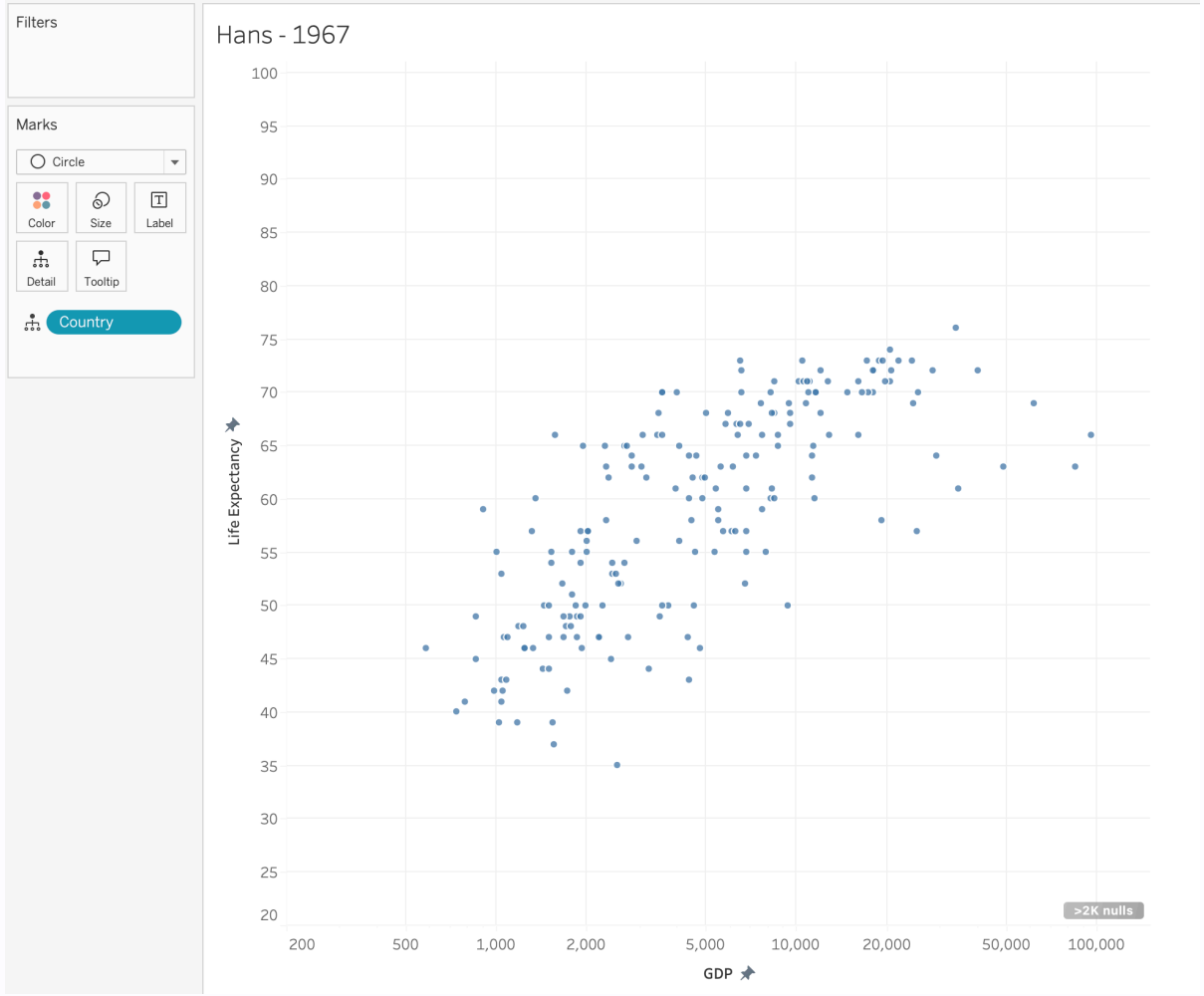
Region

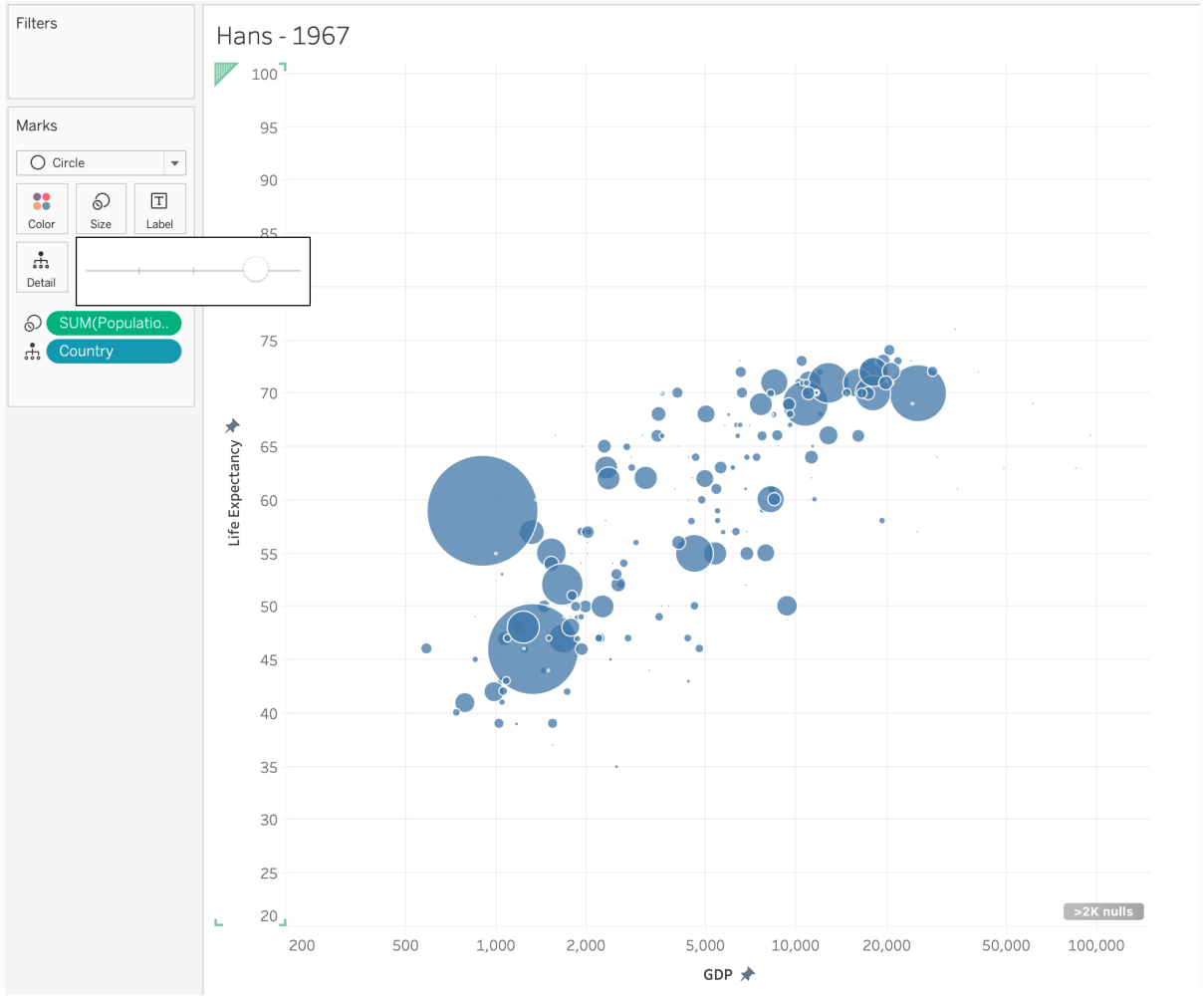
SUM(Populatio..

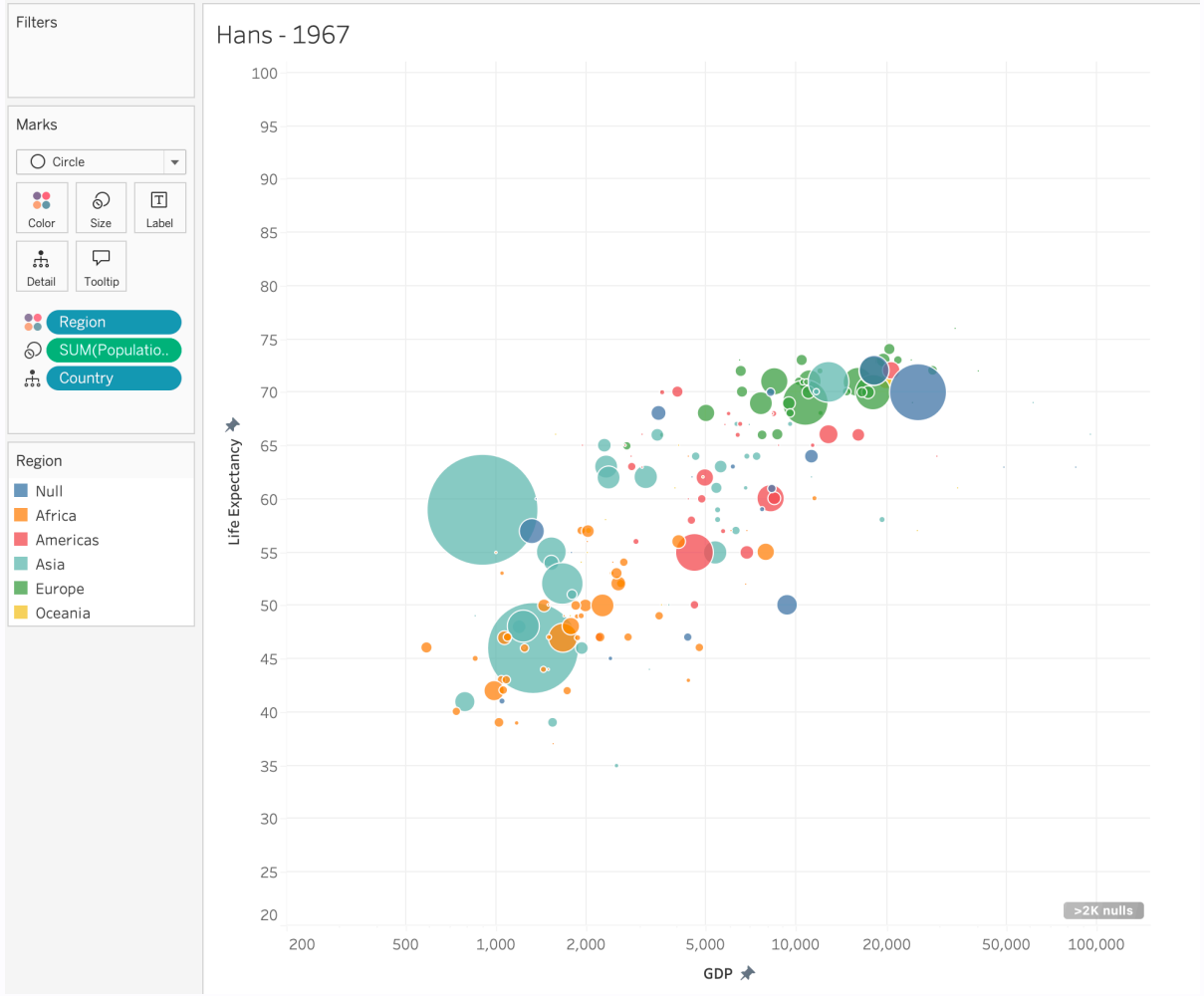
Country

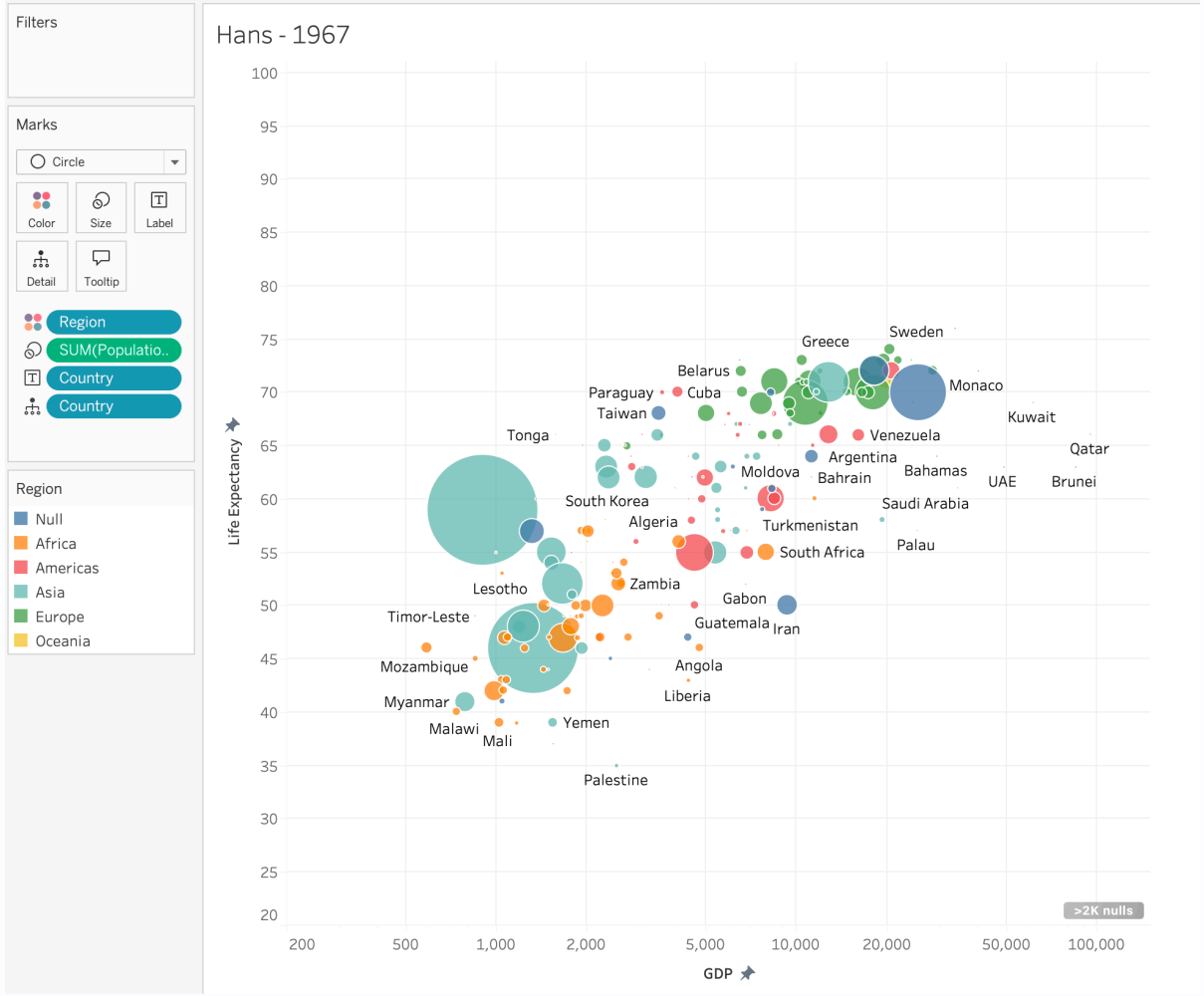
Country

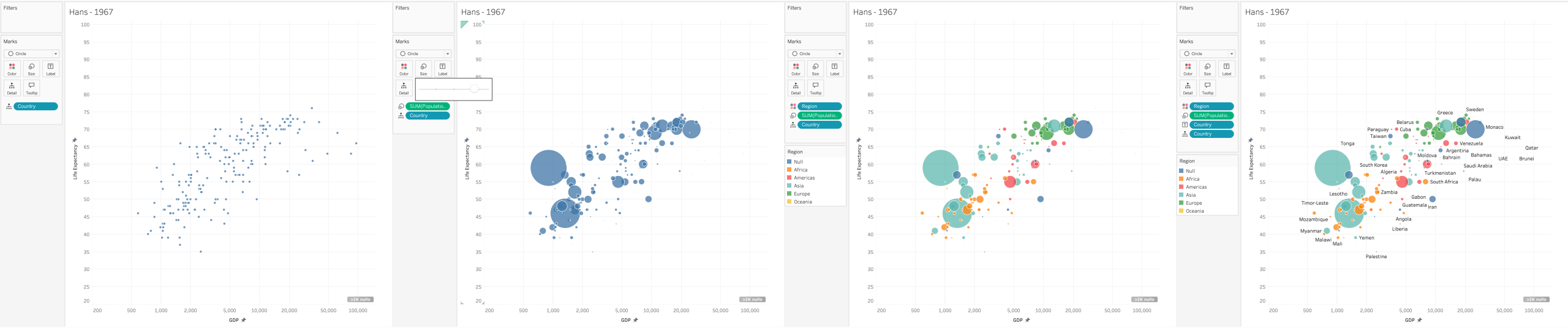



Filters
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Color
Size
Label
Detail
Tooltip
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Hans - 1967
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Life Expectancy
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GDP
>2K nulls


Filters
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SUM(Populatio..
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Life Expectancy
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20,000
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100,000
GDP
>2K nulls


Filters
Marks
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Label
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Hans - 1967
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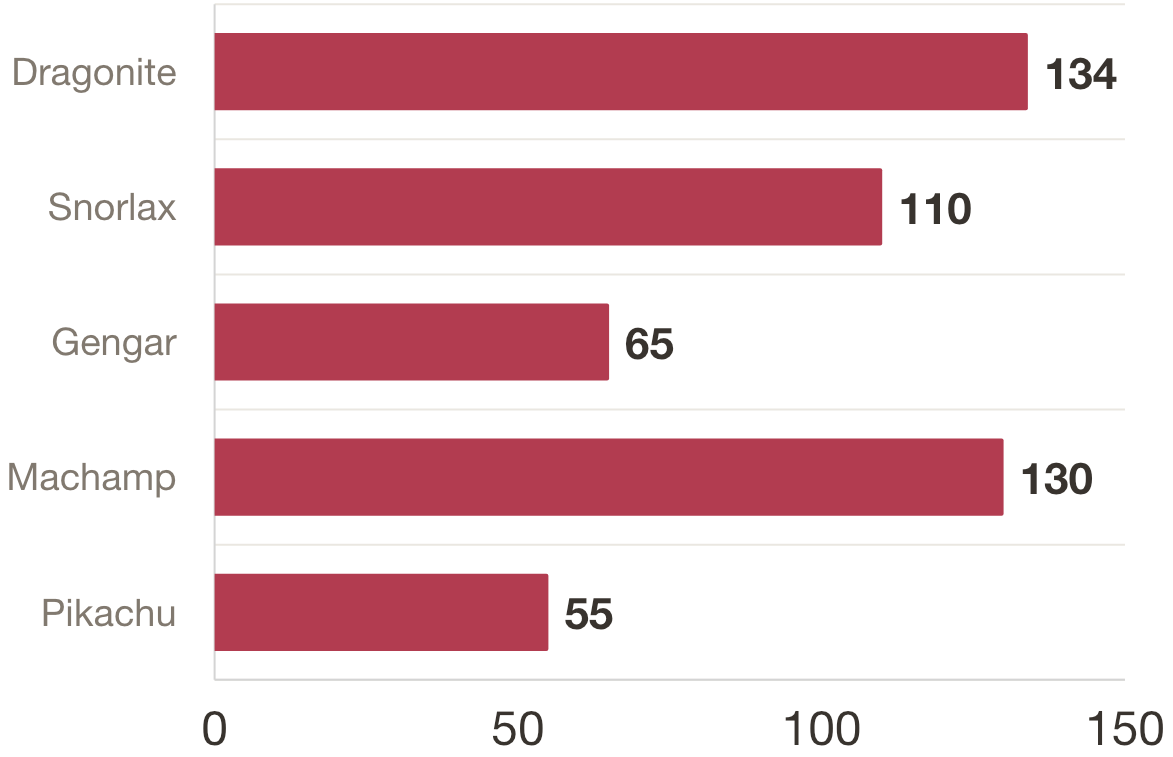

Filters
Marks
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Hans - 1967
Life Expectancy
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Greece
Sweden
Belarus
Monaco
Paraguay
Cuba
Taiwan
Kuwait
Tonga
Venezuela
Qatar
Argentina
Moldova
Bahrain
Bahamas
UAE
Brunei
South Korea
Saudi Arabia
Algeria
Turkmenistan
Palau
South Africa
Zambia
Lesotho
Gabon
Timor-Leste
Guatemala
Iran
Mozambique
Angola
Liberia
Myanmar
Yemen
Malawi
Mali
Palestine
>2K nulls


Filters
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Country
Hans - 1967
Life Expectancy
GDP
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SUM(Populatio..
Region
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Sweden
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Belarus
Monaco
Paraguay
Cuba
Taiwan
Kuwait
Tonga
Venezuela
Argentina
Qatar
Moldova
Bahrain
Bahamas
UAE
Brunei
South Korea
Saudi Arabia
Algeria
Turkmenistan
South Africa
Palau
Lesotho
Zambia
Gabon
Timor-Leste
Guatemala
Iran
Mozambique
Angola
Myanmar
Liberia
Malawi
Yemen
Mali
Palestine

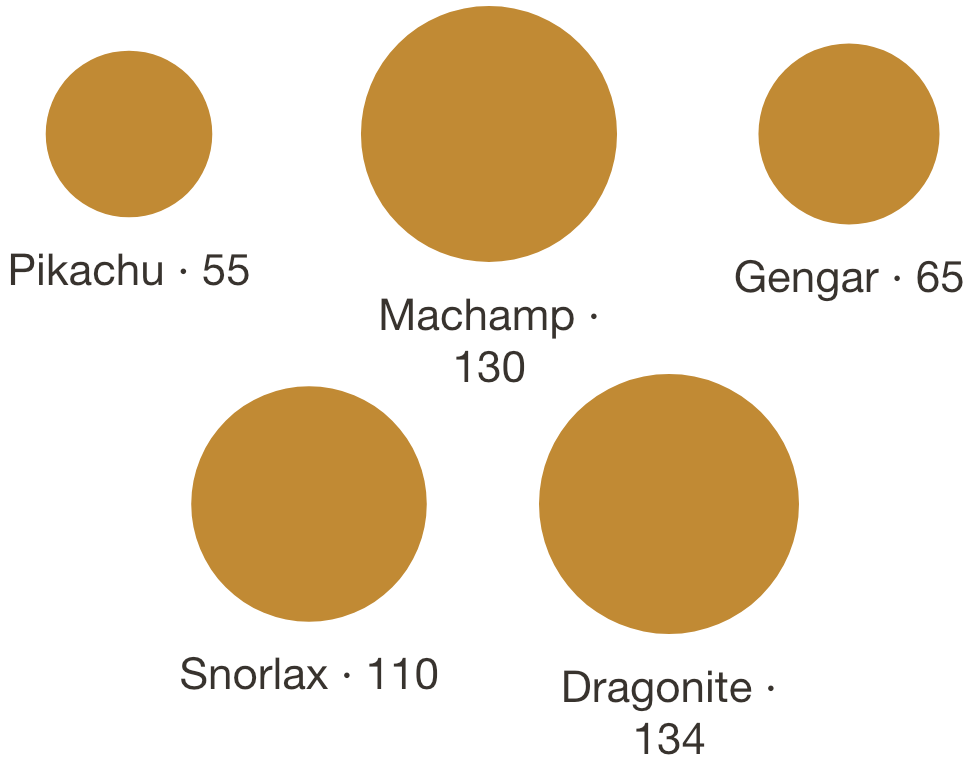
# Which comparison is easier?

Same Pokémon. Same Attack values.

## POSITION + LENGTH




## AREA

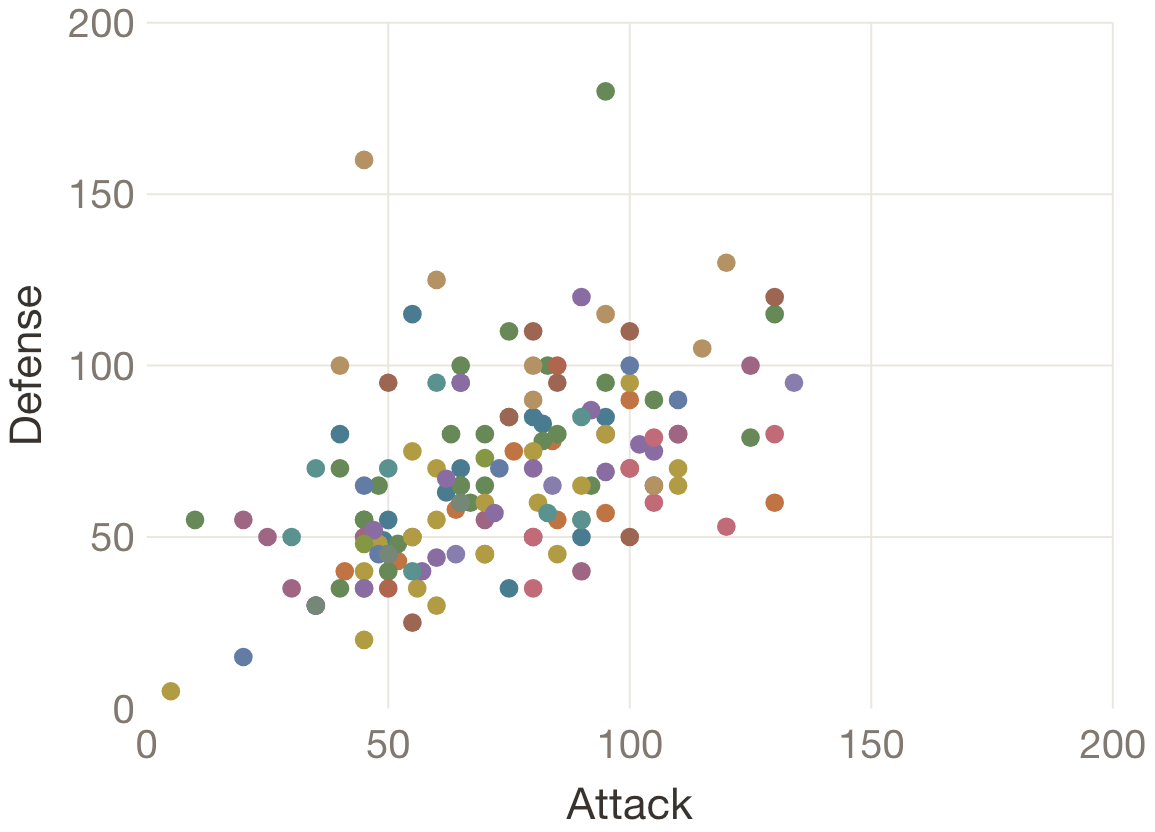



5 selected species · base Attack

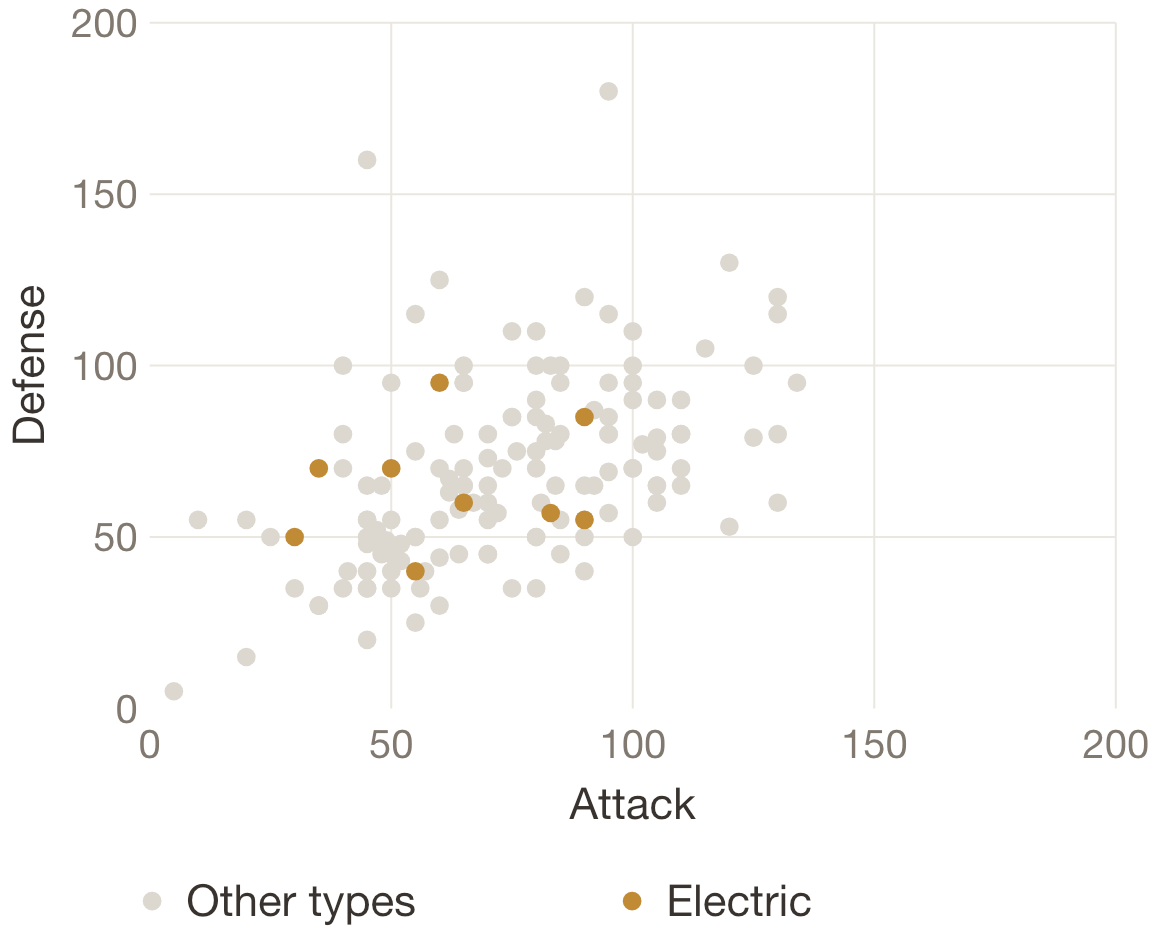
# What catches your eye first?

Color everything — or direct attention?

ALL TYPES



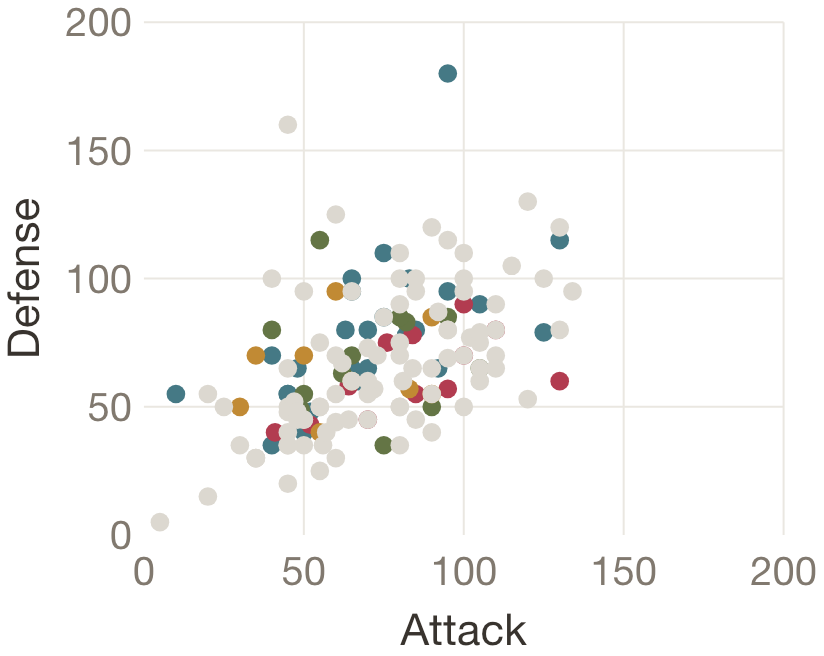

ELECTRIC IN FOCUS



# Color needs a job.

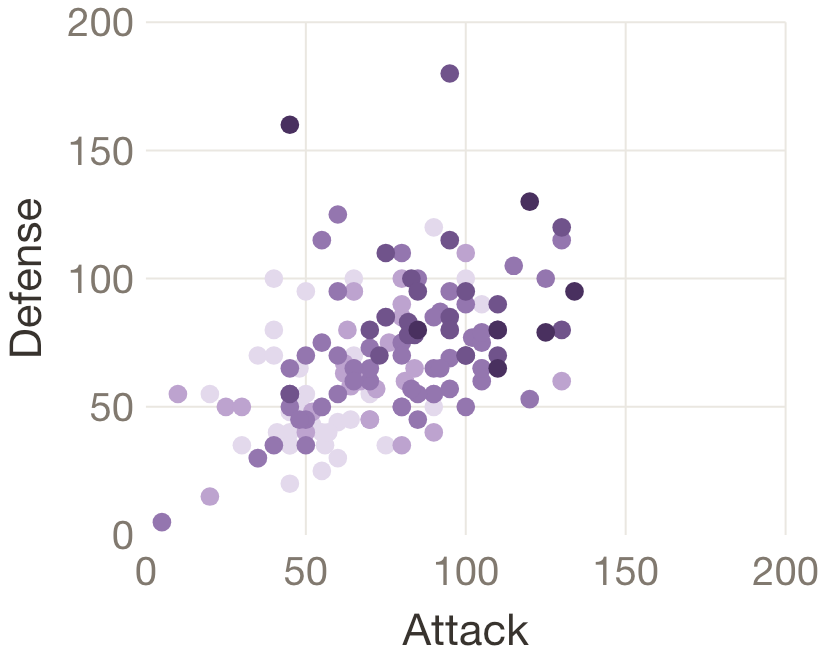
Different kinds of data need different color logic.

## TYPE



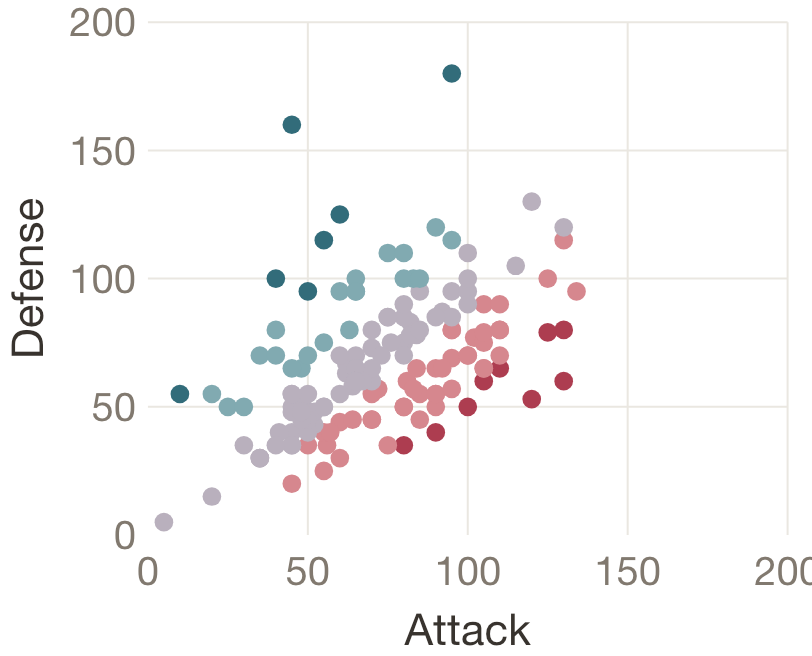

- Water
- Fire
- Grass
- Electric
- Other

## WEIGHT




- 0–<10 kg
- 10–<30 kg
- 30–<70 kg
- 70–<150 kg
- 150+ kg

## DEFENSE – ATTACK



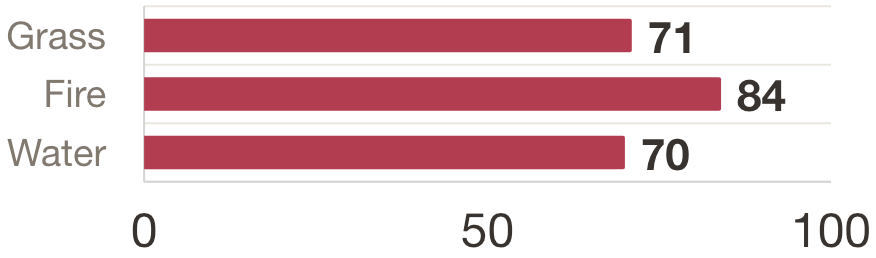

- ≤–41
- –40…–11
- –10…10
- 11…40
- ≥41

# Layout creates relationships.

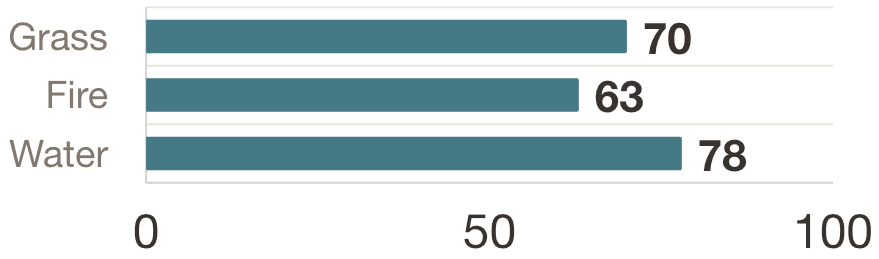
Same values. Different alignment.

## SCATTERED

### Attack

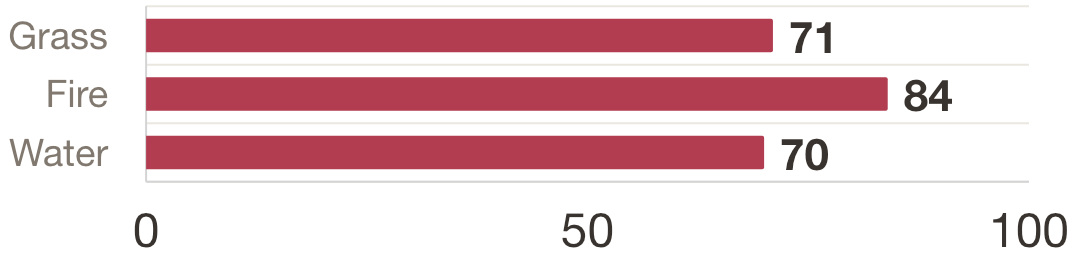



### Defense

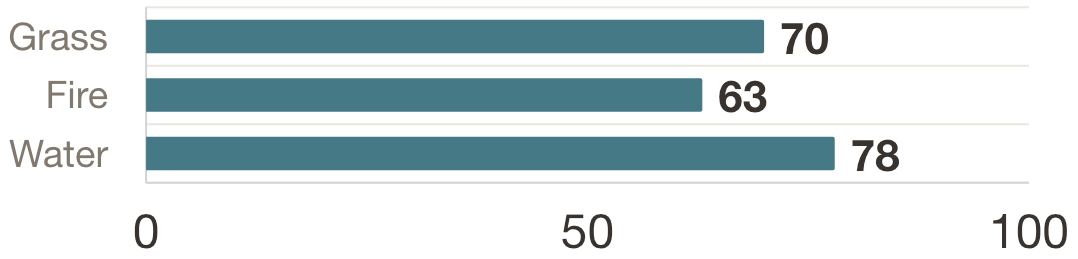



## ALIGNED

### Attack



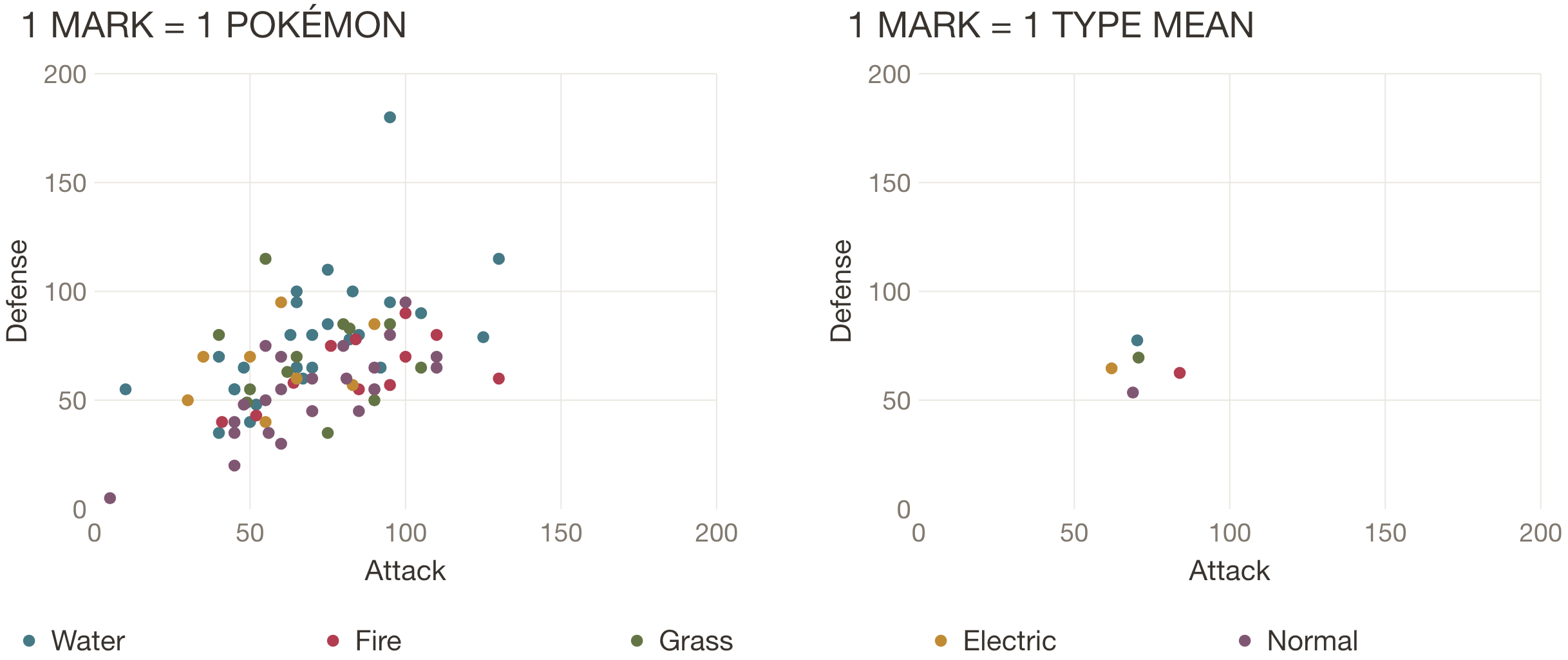

### Defense




Mean base stats · primary type · rounded to whole numbers

# What does one mark represent?

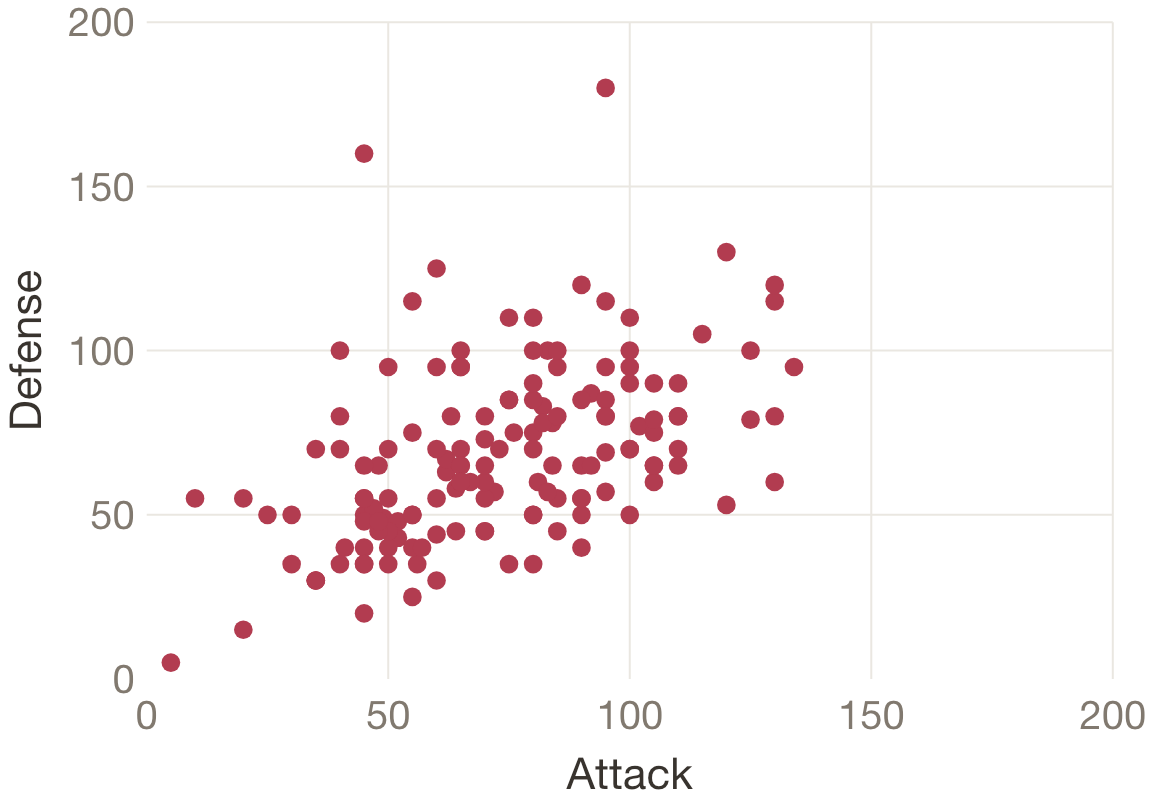
Aggregation changes the story.



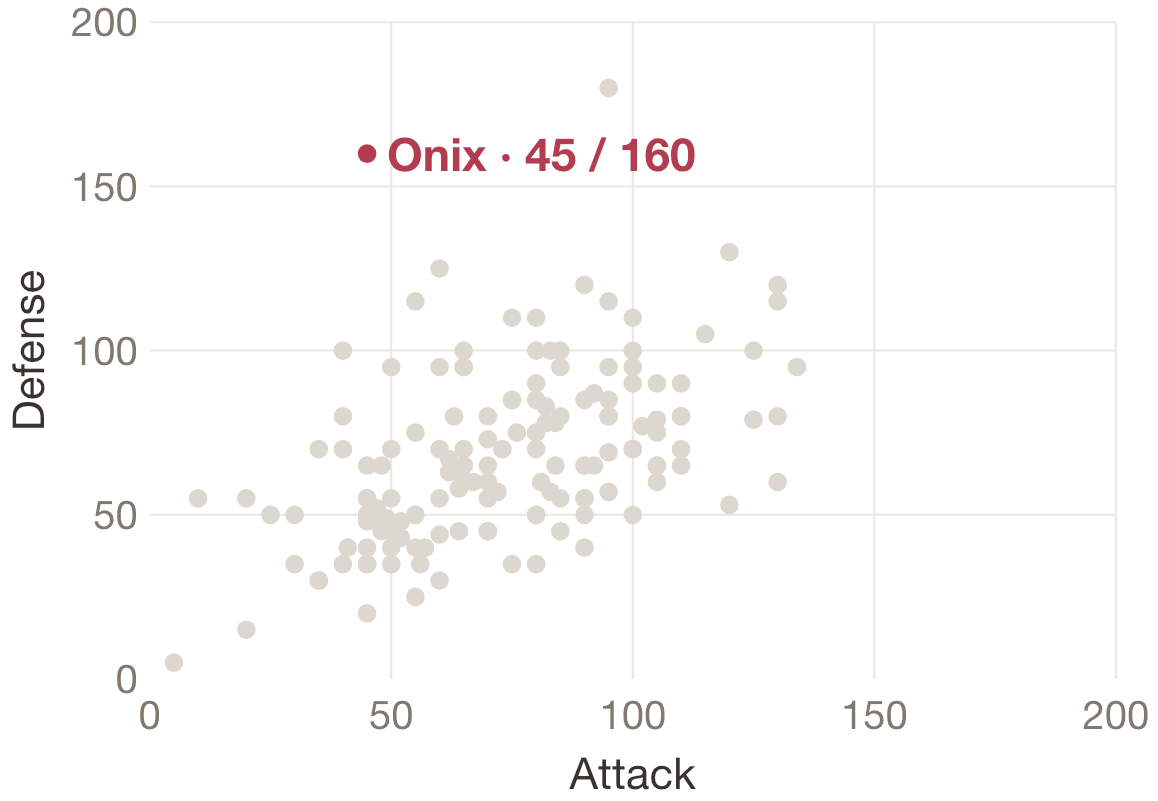
# Help the reader find the point.

A title is a chance to say something.

## Pokémon stats




## Onix: high Defense, low Attack



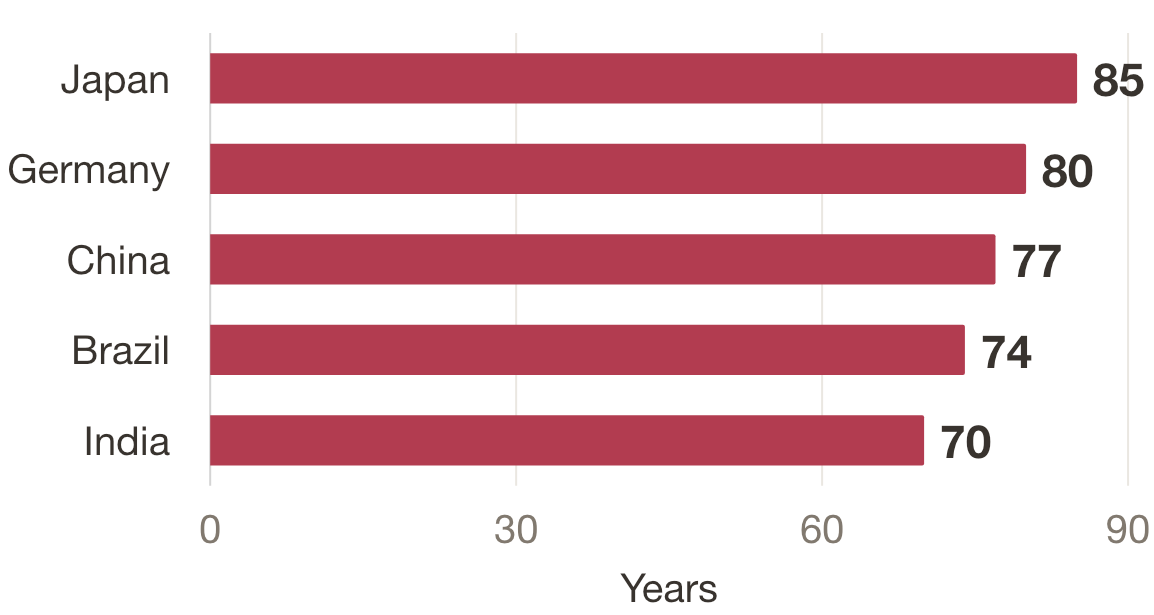

ONE CLEAR CLAIM. ONE USEFUL LABEL.

# Tableau chart types: comparisons

Life expectancy in five countries (2020)

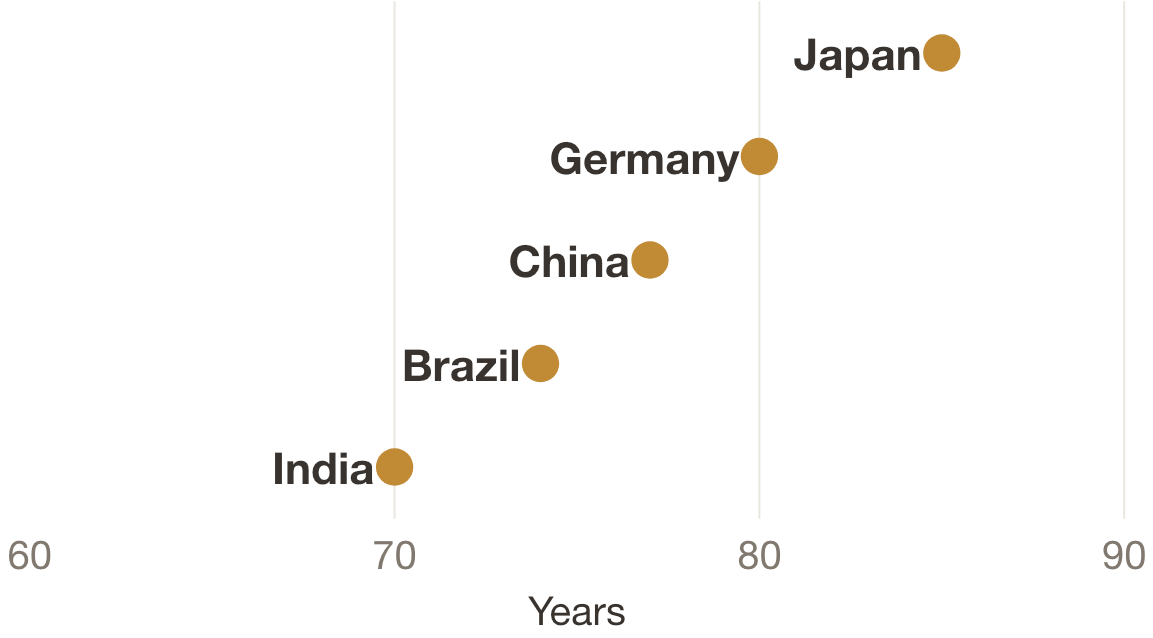
## Bar chart

Lengths compare values. Start the scale at zero.




## Dot plot

Positions make small differences easier to see.



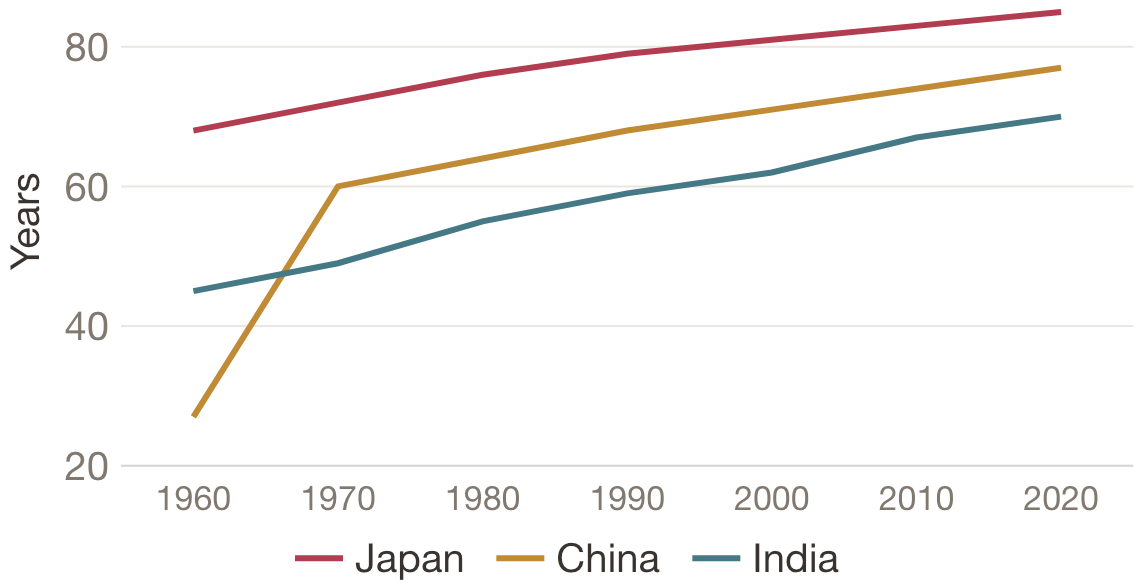

Select fields, then open Show Me. The options depend on your fields.

# Tableau chart types: change over time

A line shows a trajectory. An area also emphasizes magnitude.

## Line chart

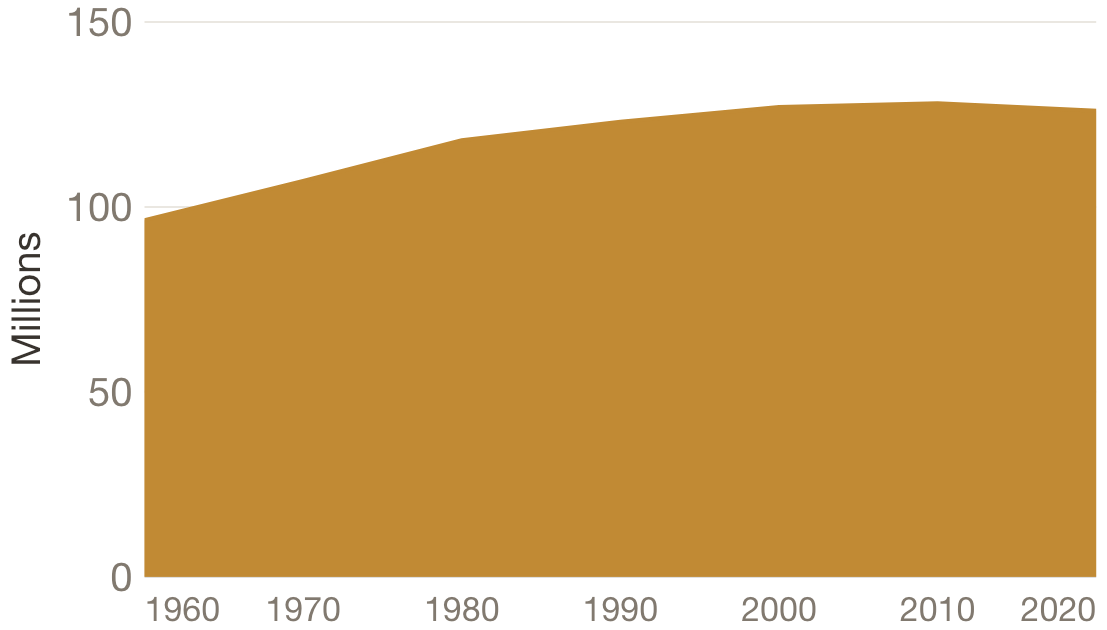
Life expectancy in Japan, China and India




Line: compare trends over time.

## Area chart

Japan's population, in millions



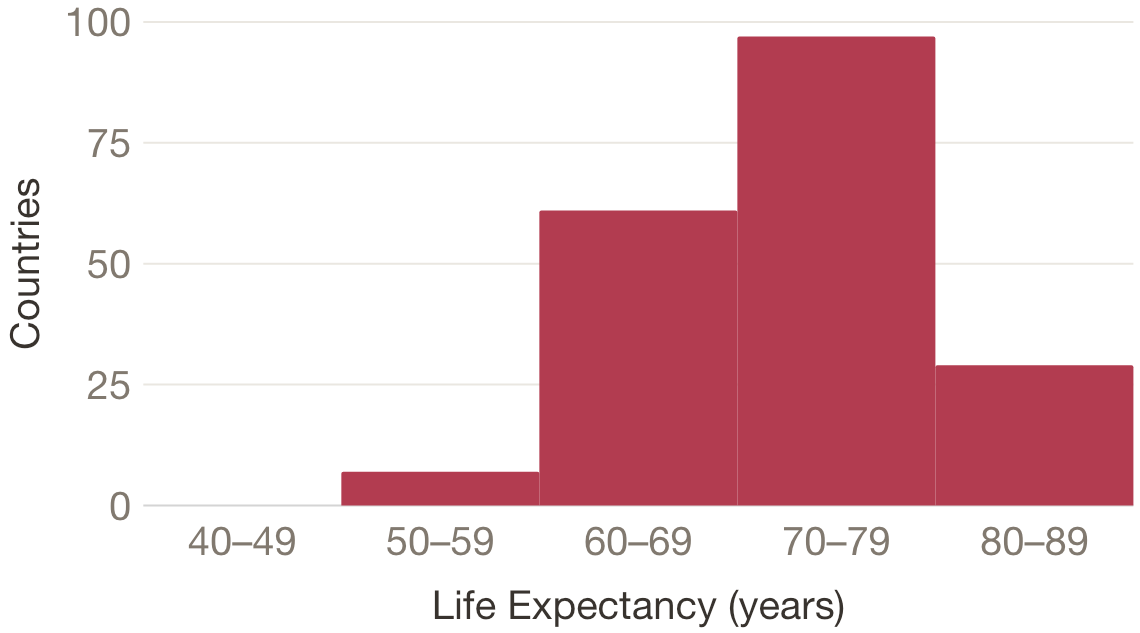

Area: use a meaningful zero baseline.

# Tableau chart types: distributions

What is typical, and how much variation is there?

## Histogram

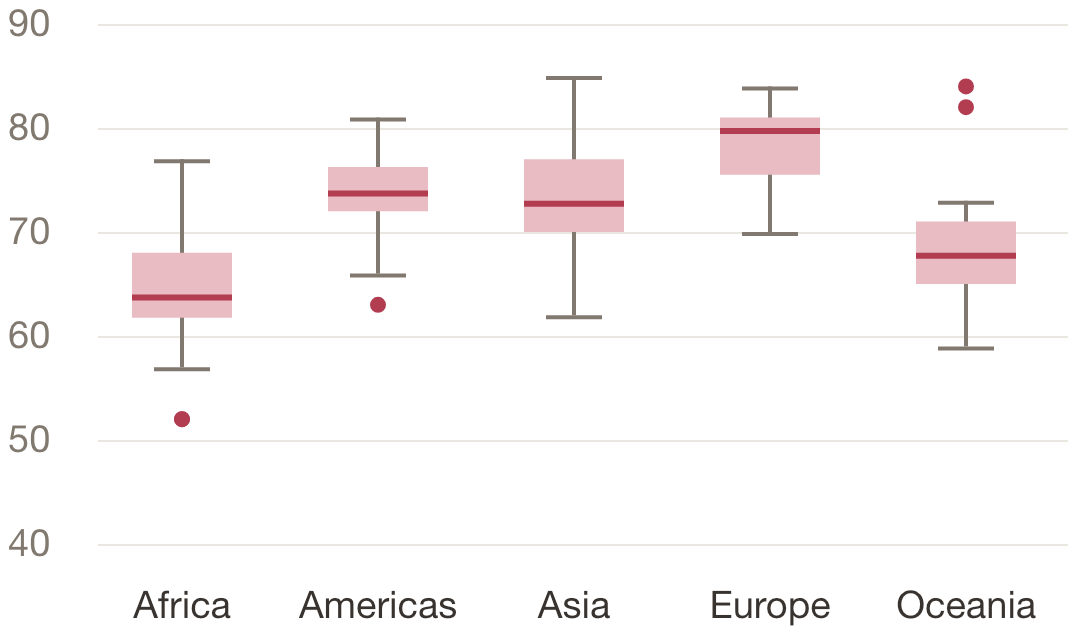
How many countries fall in each range? (2020)




A histogram counts values in numeric bins.

## Box-and-whisker plot

Country life expectancy within each region (2020)



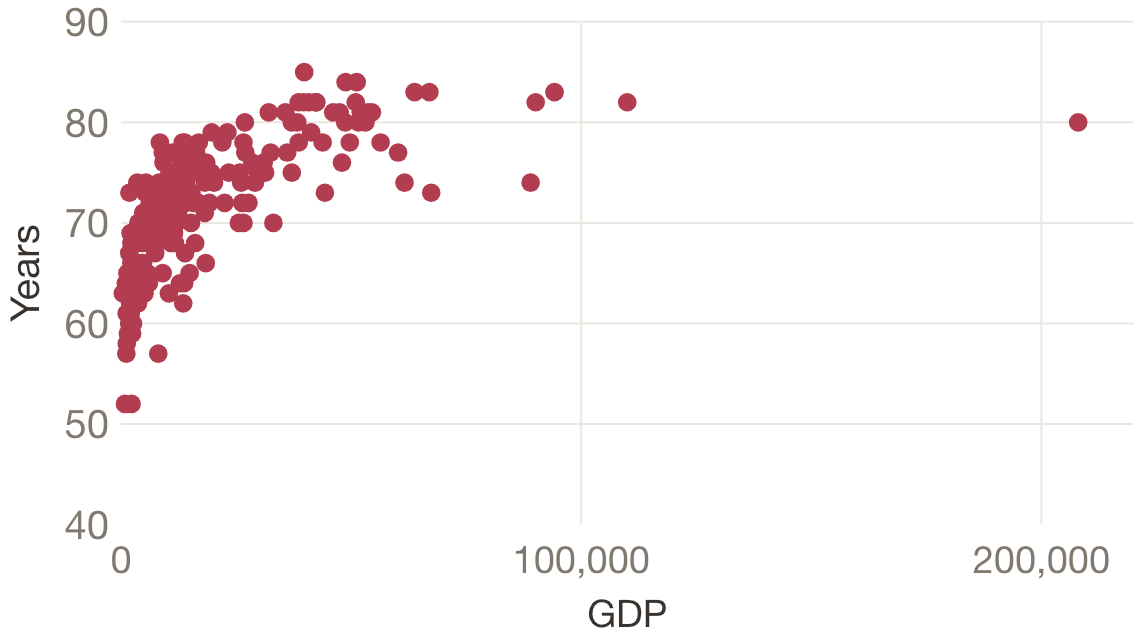

Median, middle 50%, spread and outliers.

# Tableau chart types: relationships and patterns

One pair of measures, or a matrix of repeated comparisons

## Scatter plot

GDP and life expectancy, one circle per country




Association does not establish cause.

## Highlight table

Life expectancy by country and decade

| | 1980 | 1990 | 2000 | 2010 | 2020 |
|---|---|---|---|---|---|
| Japan | 76 | 79 | 81 | 83 | 85 |
| Germany | 73 | 75 | 78 | 80 | 80 |
| China | 64 | 68 | 71 | 74 | 77 |
| Brazil | 62 | 67 | 71 | 73 | 74 |
| India | 55 | 59 | 62 | 67 | 70 |

40 50 60 70 80 90

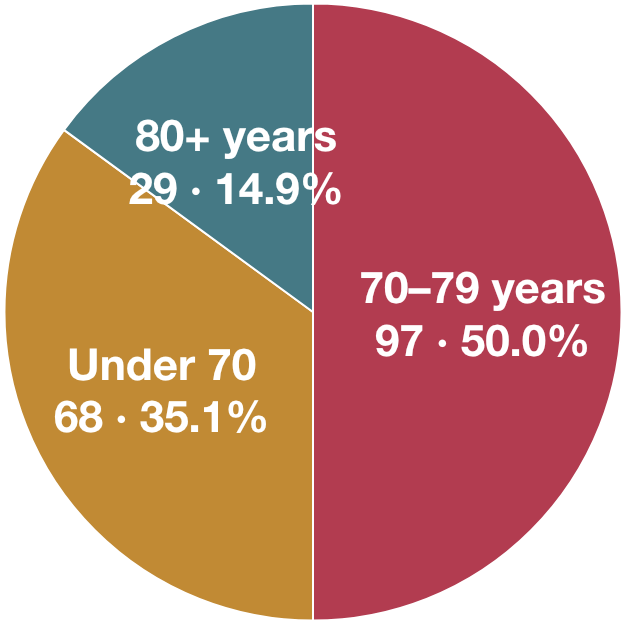
One shared color scale, in years.

# Tableau chart types: parts of a whole

How many countries fall in each life-expectancy band? (2020)

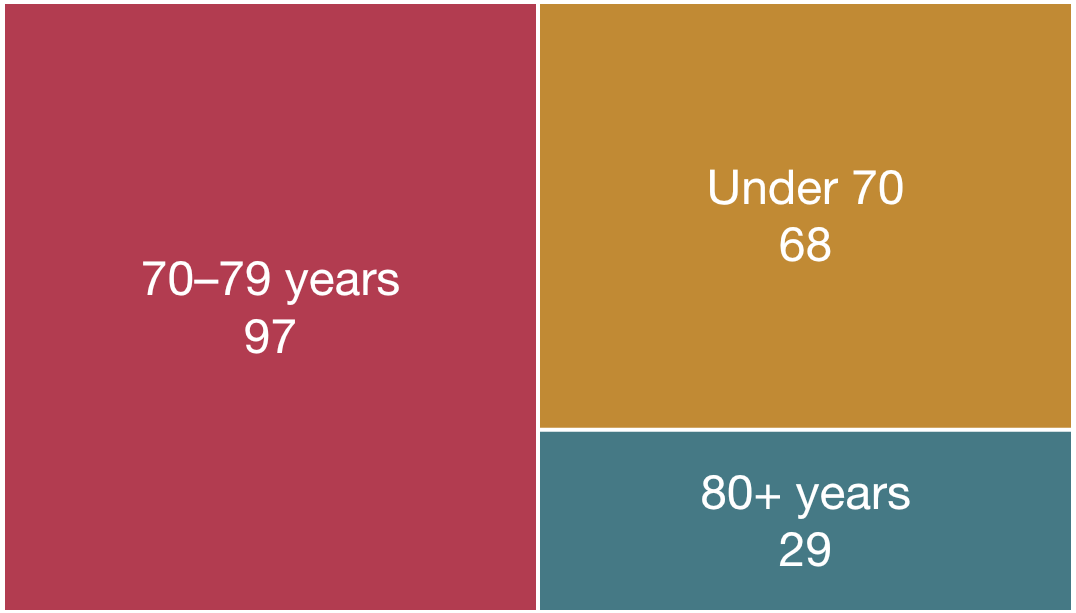
## Pie chart

Shares of a total




## Treemap

Parts sized by area




194 countries with valid life expectancy. Shares of countries, not people.

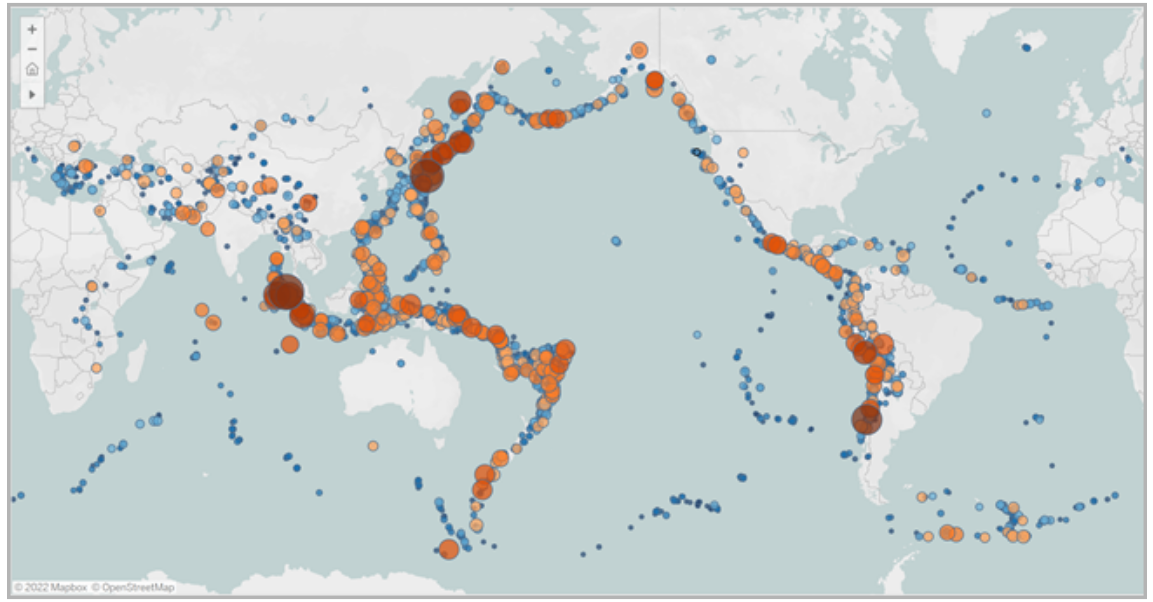
Life band → Color  COUNTD(Country) → Angle / Size

# Tableau chart types: geography

A map answers a question about location

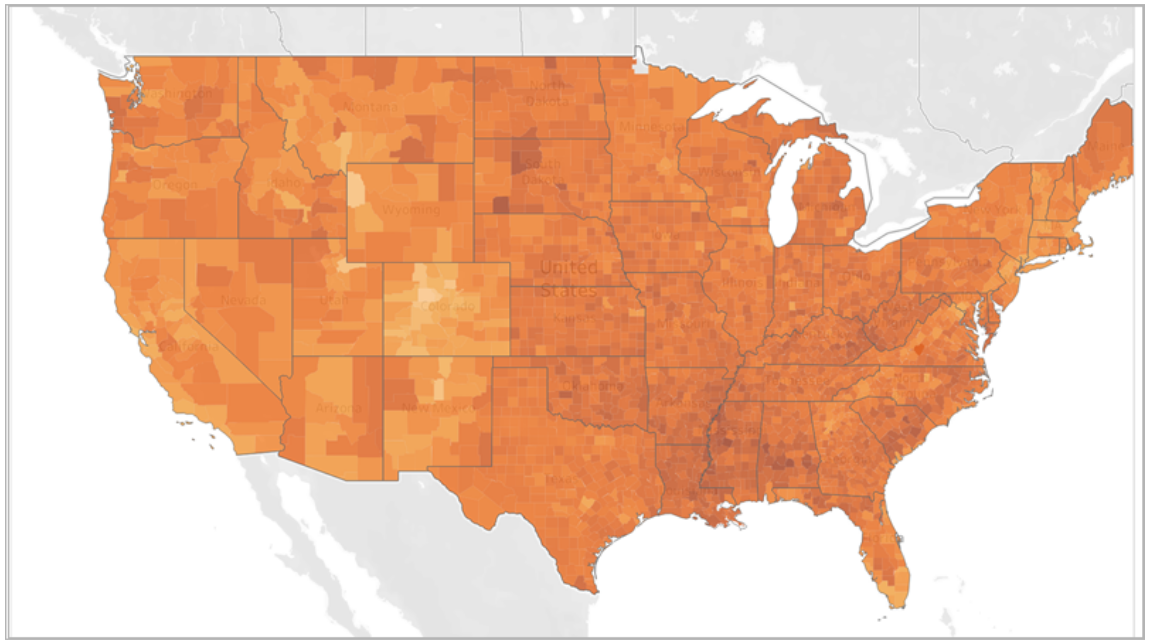
## Symbol map

Size represents a value at each location.




## Filled map

Color represents a value for each area.

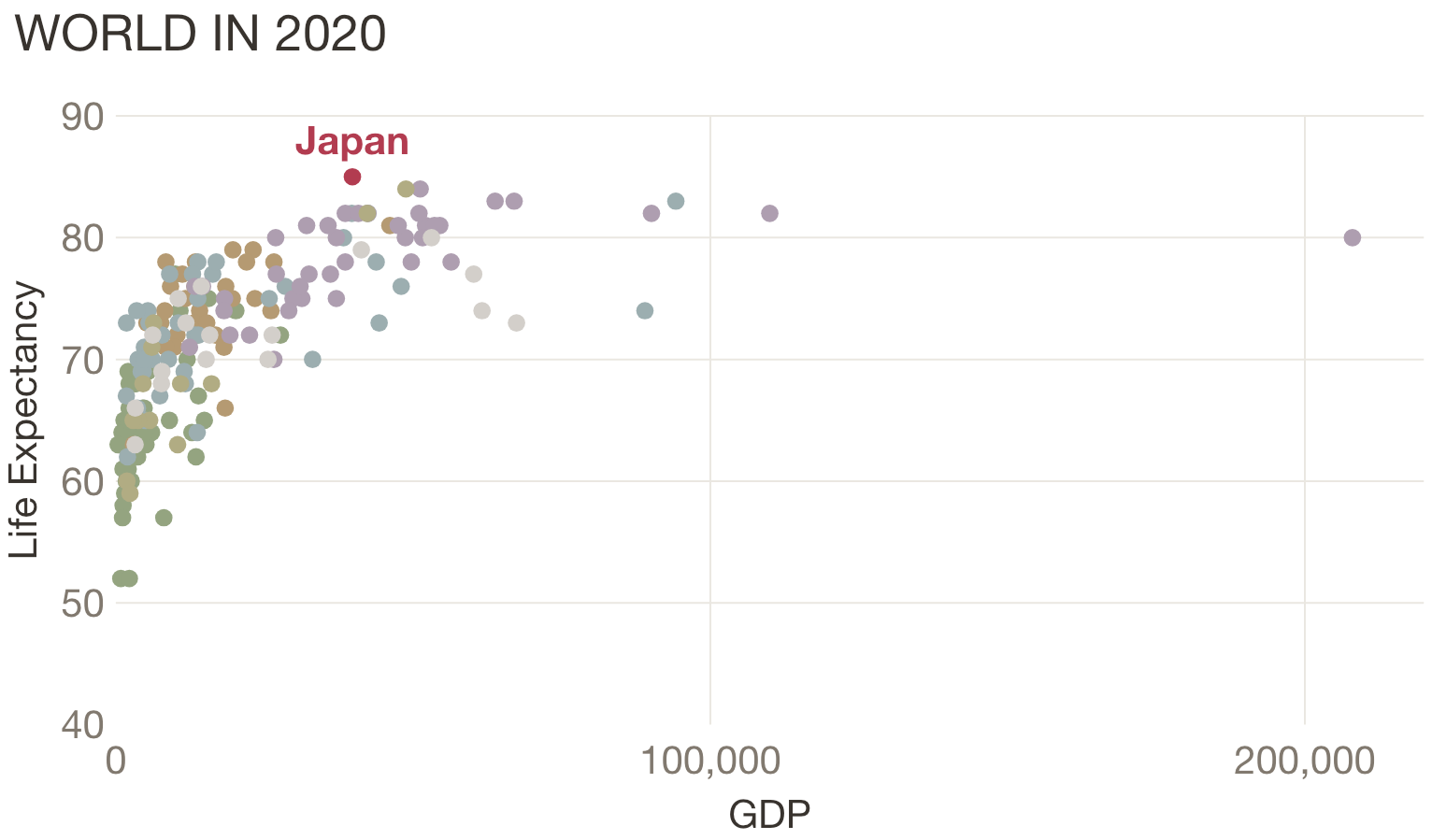


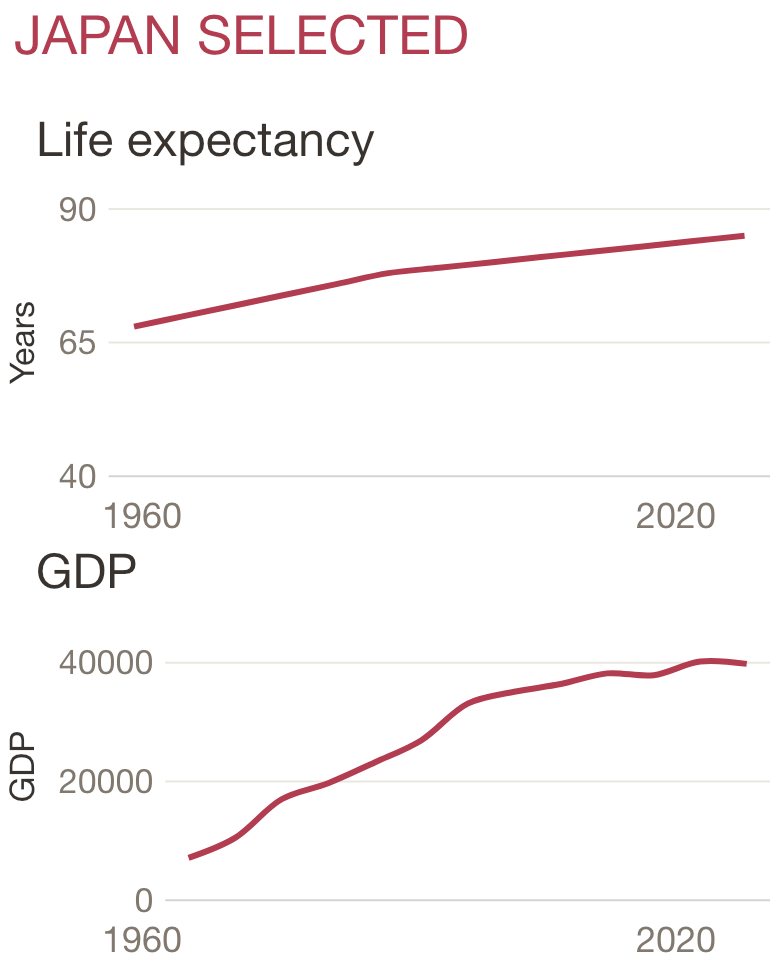

Try hans.csv: Country as a geographic field; Year = 2020; Life Expectancy on Color.

Map images: Tableau documentation examples

# Today's tip: a Tableau dashboard

Last week's Gapminder chart, connected to a country's history




Click a country in the overview. Explore its history on the right.

# The three worksheets

Right-click Year in the Data pane: Convert to Dimension

## 1 World

| | |
|---|---|
| Columns | AVG(GDP) |
| Rows | AVG(Life Expectancy) |
| Filters | Year: 2020 |

Marks: Circle

Country on Detail
Region on Color

Filters: GDP > 0
Life Expectancy > 0

## 2 Life History

| | |
|---|---|
| Columns | Year |
| Rows | AVG(Life Expectancy) |
| Filters | Year: 1960–2020 |

Marks: Line

Country on Color

Year pill menu: Continuous
Keep the full year range.

## 3 GDP History

| | |
|---|---|
| Columns | Year |
| Rows | AVG(GDP) |
| Filters | Year: 1960–2020 |

Marks: Line

Country on Color

Duplicate Life History.
Replace the measure on Rows.

Measure pill menu: Measure > Average. Leave Pages empty for this dashboard.

# The dashboard layout

A large overview and a smaller history column

1200 × 800




Tiled: horizontal container, with a vertical container on the right

## 1 Dashboard size

Turn Device Preview off. Select Default.
Dashboard pane > Size > Fixed size
Custom: Width 1200, Height 800

Drop sheets in the white canvas.
Gray space is outside the dashboard.

## 2 Tiled layout

World on the left.
Both histories in a vertical container on the right.

## 3 Year filter scope

Year filter menu: Apply to Worksheets
Selected Worksheets: World only
Histories keep 1960–2020.

# Filtering with the sheet’s funnel icon

World sheet > Use as Filter

## 1 Select the sheet

Click World in the dashboard.

## 2 Use as Filter

Click the funnel icon on its toolbar.

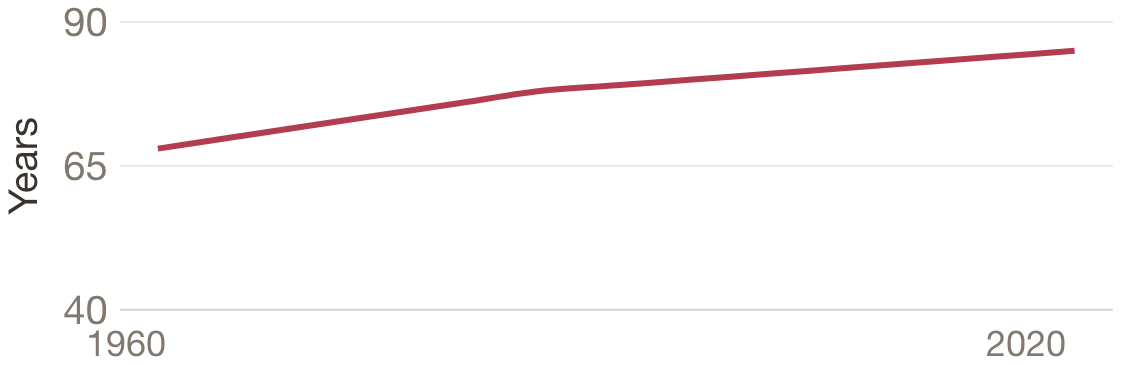
## 3 Select a country

Click Japan, then try India.

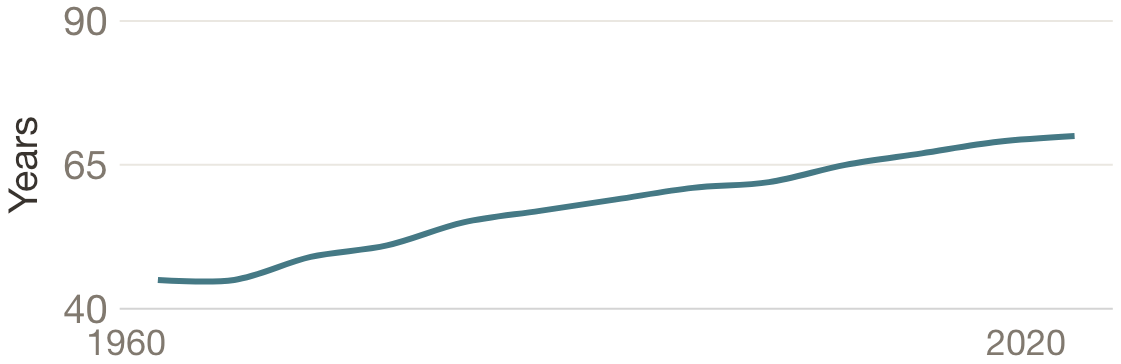
### If a history collapses to 2020:

Dashboard > Actions: edit the generated filter.
Selected Fields: Country → Country only
Clear selection: Show all values

### Click Japan




### Then click India




Life expectancy, 1960–2020

# Highlighting the selected country

Dashboard > Actions > Add Action > Highlight

Source sheet

World

Run action on

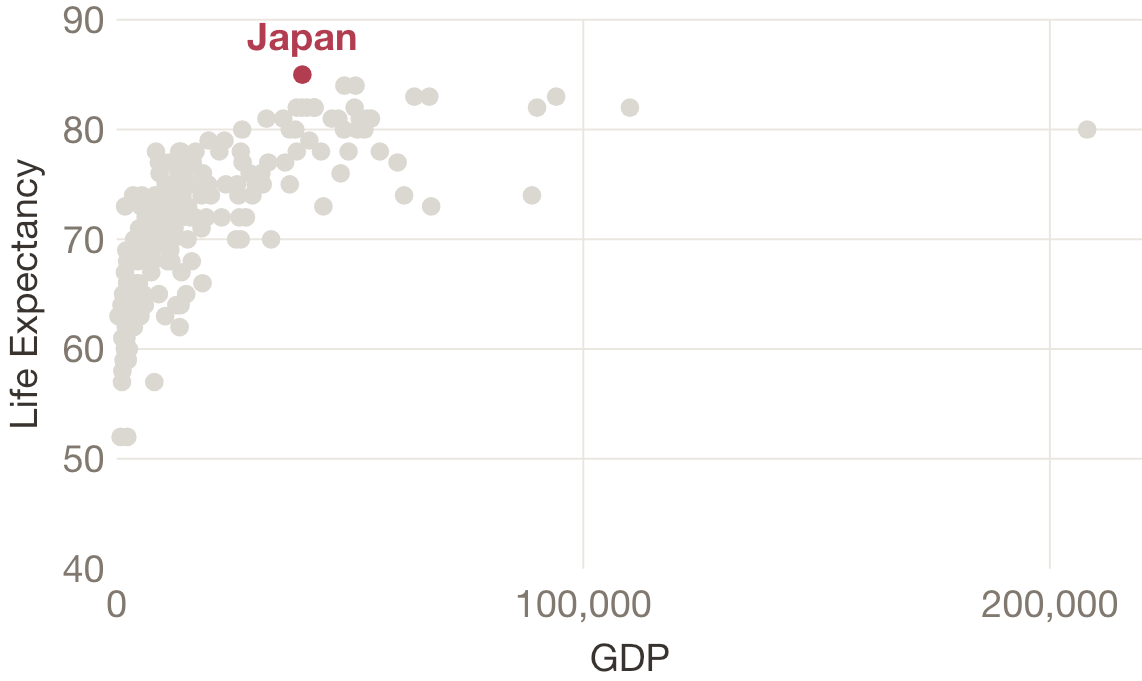
Select

Target sheet

World

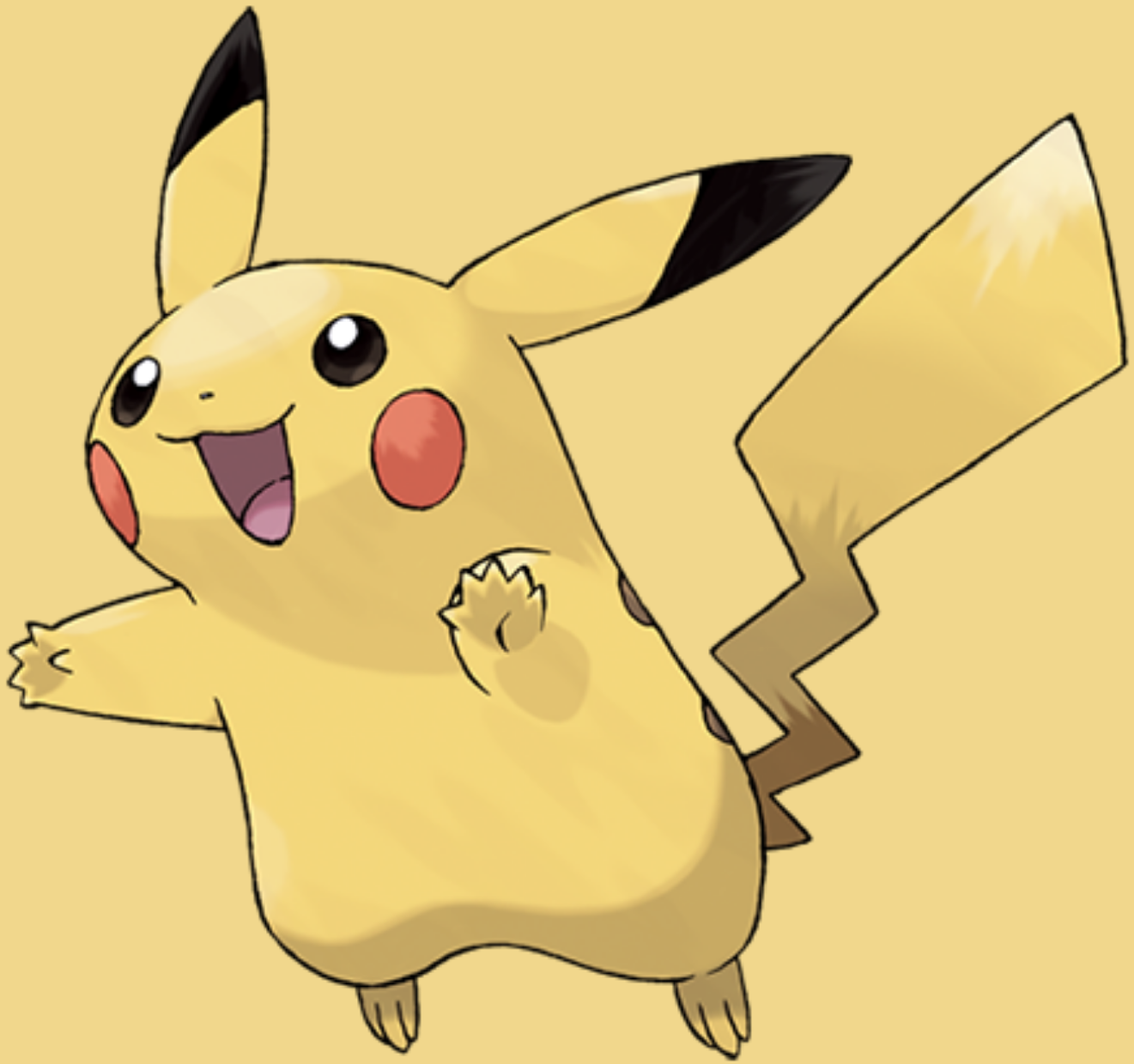
Target Highlighting

Selected Fields: Country




Click Japan: other circles fade.
Click empty space to clear.

Highlight: keep other countries visible. Filter: show only the selected country.



# Pokémon Challenge

One dataset. Your question. No Pokémon knowledge needed.

## 15 min  Solo

Choose a question. Make one chart in Tableau.
Give it a title that communicates your finding.

## Groups of 4  Share

Everyone explains their finding and one visual choice.
Choose one chart for its insight and clarity.

## One group submission

Submit names, title, finding and why you chose it.
Include one screenshot. A second is optional.

CSV + variable guide

Submit group work

Results / show and tell

# A first chart with Pokémon images

Compare Attack for five Pokémon

01 Open updated pokemon-review.csv. Open a new worksheet.

02 Name to Filters: select the five shown.

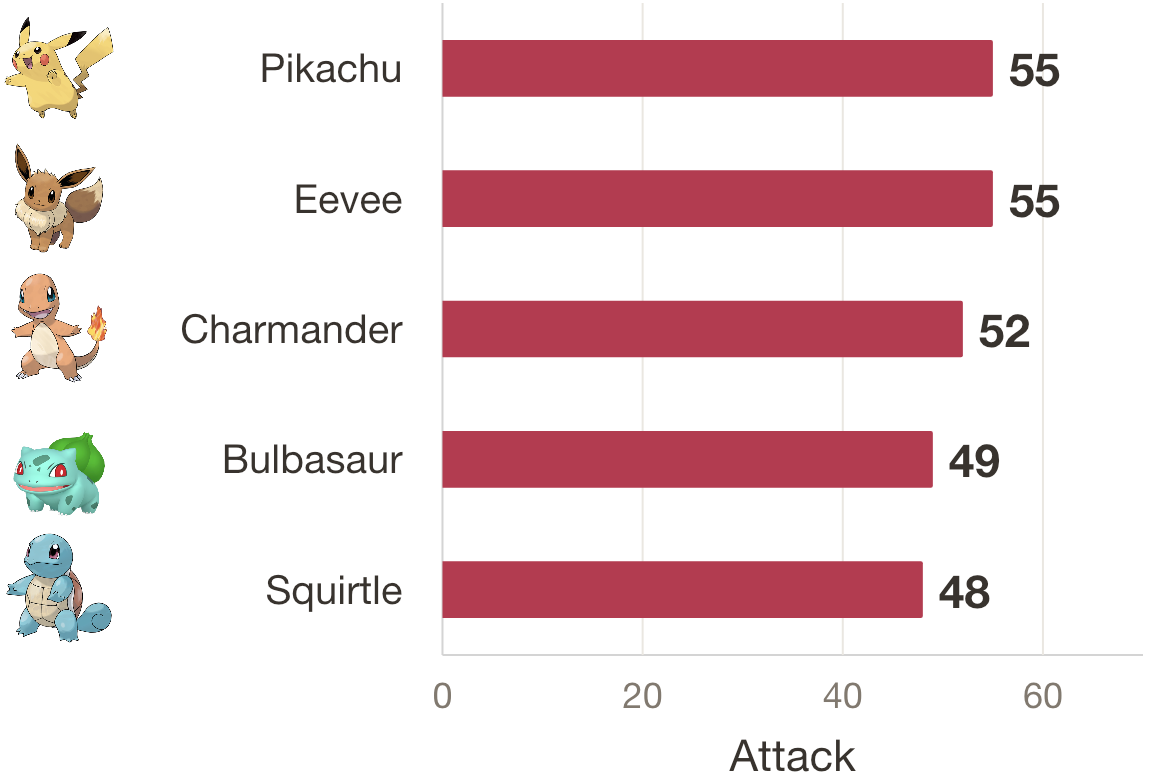
03 Right-click Image_URL in the Data pane: Image Role > URL.

04 Rows: Image_URL first, then Name.

05 Columns: Attack > Measure > Average. Marks: Bar.

06 Image_URL > Sort: Descending, Field, Attack, Average. Marks > Label: show labels.

07 Standard view. Widen the image header. Format > Cell Size > Taller. Axis starts at 0.




Images are row headers.

# Homework: a filtered dashboard

Your data. Existing or new.

01 2+ sheets in one dashboard

02 At least one working filter

03 Publish on Tableau Public

04 Submit the project link, 1 screenshot (2nd optional), a short explanation, and your data/source URL

How-to guide

Submit individually through the usual homework form.